

Enhancing Crowd Accuracy in AI-Assisted Judgmental Forecasting: Theory and Experimental Evidence

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Problem definition: Forecasts for key variables in a variety of operational decisions may be generated by human experts or by artificial intelligence (AI) technologies. A decision maker can benefit from the distinct advantages that each source may offer by providing AI assistance to the experts, allowing them to augment the information contained in the AI forecast by incorporating their own knowledge about the uncertain target variable. When multiple experts are available, accuracy can be further improved by utilizing the *wisdom of crowds*, forming a consensus by averaging each of their AI-assisted forecasts. However, the potential accuracy of a crowd of AI-assisted forecasters may be limited by two structural characteristics. First, because the AI assistance is valuable to each expert at an individual level, the opinion of the AI can end up being overrepresented in the crowd’s consensus. Second, the experts may fail to appropriately utilize the AI assistance when forming their forecasts by underemphasizing the information it provides.

Methodology/results: Using a stylized Bayesian model of information aggregation and a two-step forecast elicitation procedure, we develop a prescriptive method that can recover the most accurate consensus forecast given all information collectively observed by the AI and every expert in the crowd. This procedure works by *pivoting* the crowd’s average AI-assisted forecast either toward or away from the crowd’s average initial forecast, elicited before the forecasters have observed the AI advice. We test the predictive performance of a data-driven pivoting method in three laboratory experiments. Across 3,188 participants and three AI accuracy treatments, we find that the pivoted forecasts match, and in many cases outperform, the AI-assisted crowd, the AI advice itself, and the unassisted crowd of forecasters.

Managerial implications: Firms can use the proposed forecasting method to improve human-AI collaborative forecasting accuracy by identifying and adjusting for the net bias in the crowd’s AI-assisted forecast.

Key words: Human-AI Collaboration, Demand Forecasting, Artificial Intelligence, Wisdom of the Crowd.

1 Introduction

Generating accurate forecasts of key variables is an important task in many managerial decision problems. For example, demand forecasts serve as crucial inputs to operational decisions such

as production planning, capacity commitments, and staffing schedules (Harvey and Fischer 1997, Fildes et al. 2009, Fisher and Vaidyanathan 2014). Because the firm must commit to many of these decisions before the actual demand is realized, forecast errors can propagate through the supply chain in the form of excess inventory, capacity bottlenecks, or delayed fulfillment of orders. As a result, even modest improvements in demand forecasting accuracy can yield substantial increases in profit (Kesavan et al. 2025).

Consider the example of a Chief Operating Officer (COO) of a semiconductor manufacturer who relies on demand forecasts to make production and capacity decisions (Peng et al. 2012). Semiconductor markets are characterized by periods of cyclic demand and changing macroeconomic conditions, which makes it inherently challenging to predict near-term demand. If demand is overestimated, the firm may commit fabrication capacity and raw materials to chips that customers no longer need; if demand is underestimated, it may face delayed fulfillment and lost sales in downstream markets. Because demand for semiconductors can be influenced by a range of technological, economic, and geopolitical factors, the COO may benefit from eliciting forecasts from one or more subject-matter experts in these areas (Gaur et al. 2007). An engineering manager may track emerging trends in technical chip requirements, a policy analyst may anticipate downstream changes in industrial policy that influence market dynamics, while a geopolitical expert may evaluate trade restrictions that would affect global demand for semiconductors. When multiple experts are available, a large body of work shows that accuracy of predictions can often be improved by averaging individual forecasts obtained from the experts—a phenomenon known as the *wisdom of crowds* (Armstrong 2001, Surowiecki 2005). Intuitively, averaging opinions elicited from a crowd of forecasters enables the decision maker (DM) to combine the unique information available to each expert, while allowing idiosyncratic errors in each of their forecasts to cancel each other out.

Artificial intelligence (AI), machine learning, and statistical prediction models can be useful resources for these experts in improving their forecast accuracy. Firms across multiple sectors are rapidly investing in AI-enabled business forecasting capabilities; Gartner (2025) predicts that 70% of large organizations globally will adopt AI-based supply chain forecasting to predict future demand by 2030. These technology-enabled forecasting tools can be both quantitative and qualitative in nature. Large language models are capable of collating unstructured and dispersed information into cohesive, natural-language opinions about the demand environment, while statistical methods can leverage big data to produce a forecast with minimal predictive error—and both types of algorithmic output may add value to the forecasting process. Algorithmic output from any of these approaches, which we will refer to broadly as *AI advice* in this paper, can in general improve the predictive accuracy of judge-advisor systems when provided to experts as external assistance (Dai and Swaminathan 2025). The short- and long-term performance impact of AI advice is linked

to the type and nature of the assistance being provided (Bastani et al. 2025, Poulidis et al. 2025a,b), which makes the design of optimal algorithmic advisor systems an important research agenda.

In many demand forecasting tasks, humans and machines can work together in a complementary manner. On one hand, human experts may possess information or context about the demand environment that is not accessible to an algorithm. On the other hand, machines have the ability to process very large amounts of data which are beyond the cognitive capacity of a human. This complementarity in skill is often leveraged by allowing a human expert to make a *judgmental adjustment* to the algorithm’s recommendation (Sanders and Ritzman 2001). Returning to the earlier example, a geopolitical expert may find it useful to study the results of a time series forecasting model trained on historical demand, as well as to review potentially relevant background information about the industry that has been gathered by a generative AI tool. At the same time, the expert may also have some unique perspective or be privy to privileged communications about developing geopolitical alliances and global trade deals that would affect near-term demand for semiconductor chips—knowledge which would typically not be reflected in an algorithm’s prediction. This notion of *private information* that resides exclusively with humans leads to complementarity between human experts and well-trained algorithms (Balakrishnan et al. 2025).

However, in spite of the improvement in individual forecasting performance that can be tapped into by providing the experts with AI advice, its value may be diminished or, in certain cases, even negative, when the opinions of multiple experts are combined. Loosely speaking, when AI advice is useful but available to all of the experts, each individual forecaster will incorporate it into their own prediction of uncertain demand and, as a result, the opinion of the AI may receive excessive weight in the crowd consensus—a phenomenon termed the *borg effect* by Fügener et al. (2021). The borg effect can be thought of as a form of shared-information bias, wherein common knowledge is over-emphasized in a crowd’s average forecast (Lichtendahl et al. 2013, Palley and Soll 2019). For instance, Doshi and Hauser (2024) show that in qualitative tasks such as creative writing, access to generative AI enhances creativity in individual writers but also suppresses the collective diversity of ideas within the group. These results highlight a danger that, in a crowd forecasting context, there may be inconspicuous drawbacks to the provision of AI advice. While it should tend to improve the accuracy of individual demand estimates, its effects on the crowd’s collective wisdom are complicated—and less explored in quantitative forecasting environments.

Of course, when considering forecasts produced by a human expert, the DM must also bear in mind the systematic behavioral tendencies that may affect their judgments (Croson et al. 2013, Donohue et al. 2018). There are numerous ways in which individuals may fail to appropriately make use of advice more generally (Davis 2018), and when advice specifically comes from an algorithm, humans show a tendency to underutilize its opinion—a phenomenon known as *algorithm aversion*

(Dietvorst et al. 2015). Kesavan and Kushwaha (2020) show field evidence from inventory decisions that the provision of discretionary judgmental adjustment to managers may even backfire upon the firm in terms of realized profit. Thus, while AI advice can help individuals make more accurate demand forecasts, behavioral frictions could prevent experts from using it to its full capacity—even when the algorithm demonstrably improves the quality of output (Wang et al. 2025).

In this paper, we propose a stylized Bayesian model of information to understand both borg effect and algorithm aversion when multiple experts utilize AI advice in a forecasting task. We show that these two phenomena have separately harmful but mutually opposing effects on the crowd’s average forecast. Based on this model, we develop a prescriptive adjustment that can recover the most accurate consensus prediction by optimally combining all of the information available to all forecasters in the crowd. This prescription takes the form of a *pivoted forecast* (Palley and Soll 2019), which adjusts the crowd’s average AI-assisted prediction either toward or away from the average unaided prediction. We derive a closed-form expression for the optimal pivoting factor under minimal assumptions about the information environment, and we provide a practical, data-driven procedure that estimates this factor from historical AI-assisted crowd forecasting data.

We evaluate the performance of the pivoted forecast in three controlled forecasting experiments where predictions from online participants on different tasks are collected twice—first without and then with AI advice. Across 3,188 participants and 200 tasks, we compare the pivoted forecasts to the average unaided crowd forecast, the average AI-assisted crowd forecast, and the AI advice itself. In all three information environments, the pivoted forecast yields the lowest mean squared error. The experiments exhibit forecasting environments with different mixtures of borg effect and algorithm aversion, resulting in different recommended pivot directions: toward the initial crowd forecast when AI advice is relatively poor, away from it when AI advice is highly accurate, and no adjustment when the two forces offset each other—a scenario that we call *null pivoting*. This highlights an appealing feature of our approach: the pivoting procedure is conservative and defaults to the average AI-assisted prediction when the data do not strongly support a correction.

The rest of the paper is organized as follows. In §2, we provide a brief review of the existing research in this domain and identify the contributions of this paper within the extant literature. In §3, we introduce a stylized model of information held by an individual human expert and the AI, define normative standards for how both should be combined to generate the most accurate final forecast, and present a behavioral model to describe how humans actually use the advice available to them. In §4, we demonstrate how a simple average of AI-assisted forecasts would not be an accurate estimate of the quantity of interest due to the two forms of distortion discussed above. In §5, we use the earlier results to show that the efficient forecast in fact takes the form of a pivoted estimate, where the size and direction of the pivot are determined by the amount of

borg effect and the degree of algorithm aversion in the crowd. §6 introduces a data-driven method to implement optimal pivoting in practice using historical forecasting data. §7 presents the results of our online experiments and tests the relative performance of pivoted forecasts on real data. Finally, in §8 we discuss the general applicability of the pivoting approach to forecasts elicited from AI-assisted crowds of experts, and conclude by highlighting some potential directions for future research. Proofs of all results are provided in §EC.1–EC.2 of the electronic companion.

2 Literature Review

Our work sits at the intersection of three streams of academic literature: judgmental adjustments in forecasting, behavioral aspects of human-AI collaboration, and forecast aggregation. Across these streams, the common operational problem is how a decision maker should form an accurate forecast in a data-driven fashion using relevant information that is dispersed across sources.

First, our work contributes to the literature on algorithmic demand forecasting with judgmental adjustments. Prior work has studied the use of judgmental forecasts to estimate demand uncertainty (Fildes et al. 2009, Moritz et al. 2014, Brau et al. 2023, Palley and Bansal 2025) and to assist in retail and supply chain decisions (Ferreira et al. 2016, Cohen et al. 2022). A consistent finding in this literature is that algorithmic forecasts and human judgment are often complementary: algorithms can process large-scale historical data, while human experts possess private information that is not reflected in the model (Balakrishnan et al. 2025). Using a field experiment, Kesavan et al. (2025) demonstrate that the provision of judgmental adjustments to inventory forecast inputs increased firm profitability by 4.92% on average, compared to a fully automated setup. Recent work has examined this complementarity across a variety of human-algorithm collaborative tasks, including demand and inventory planning (Chen et al. 2022, Nair and Huchzermeier 2024, Balakrishnan et al. 2025, Flicker 2026, DosSantos DiSorbo and Ferreira 2026), warehouse operations (Sun et al. 2022), medical decisions (Ibrahim et al. 2021), and outlier identification (DosSantos DiSorbo et al. 2025). Our paper contributes to this stream by studying how a manager should aggregate multiple AI-assisted human forecasts when both the AI and the human experts possess distinct but overlapping information about the target variable.

Second, our work relates to research on behavioral aspects of human-AI collaboration. The advice-taking literature shows that people often fail to use advice optimally, even when the advice is informative (Soll et al. 2022). When the advisor is an algorithm, this behavioral friction often appears as algorithm aversion: individuals underweight algorithmic recommendations, especially after observing that the algorithm can make mistakes (Dietvorst et al. 2015). In forecasting tasks, there is a real risk of algorithm-averse individuals placing less weight on the AI advice than they should, leading to their forecast accuracy falling short of its potential (Sun et al. 2022, Satopää

et al. 2024, Wang et al. 2025, Kormylo et al. 2026). Research in operations and information systems show that behavioral factors often largely determine the success of human–algorithm collaborations (Kesavan and Kushwaha 2020, Moosavi et al. 2025, Snyder et al. 2025, Wang et al. 2025). We analyze these behavioral findings from the lens of a prescriptive judgment aggregation problem.

Finally, our paper also builds on the literature on forecast aggregation and the wisdom of crowds. A large body of work shows that averaging forecasts from multiple experts can improve accuracy by combining dispersed private information and canceling idiosyncratic errors (Winkler 1981, Timmermann 2006); however, forecasts need to be combined with caution when information sources are correlated (Satopää et al. 2016, Palley and Satopää 2023). This literature shows that the predictive performance of an aggregate forecast depends critically on latent error correlation structures in the forecasting environment (Lichtendahl et al. 2013, Palley and Soll 2019, Satopää 2022, Jia et al. 2023). A recently developed prescriptive method to address this issue is known as pivoting (Palley and Soll 2019, Rilling 2024), in which the DM uses meta-information about forecasters’ beliefs to correct for shared-information bias. We extend this stream to a specific AI-assisted forecasting environment, in which the shared signal in the form of AI advice is explicitly provided to improve individual forecasters’ prediction accuracy, and a prescriptive optimal pivoting procedure is used to de-bias the average AI-assisted forecast.

Our paper addresses a forecasting problem at the confluence of these three literatures. We study how a DM should aggregate forecasts from a crowd of experts who first form their own demand forecasts and then revise them after receiving common AI advice. To the best of our knowledge, this is the first academic work to develop and experimentally test a prescriptive aggregation method for AI-assisted crowds of forecasters, that simultaneously accounts for structural over-weighting and behavioral under-weighting of AI advice under different information environments.

3 Human Forecasts and the Incorporation of AI Advice

To study how a crowd of experts utilizes AI advice, we introduce a general model based on a well-established Bayesian information framework for stochastically dependent forecasts (Winkler 1981, Chen and Zhao 2025). We consider a DM who needs to estimate the variable $\theta \in \mathbb{R}$ but does not possess any specific information about its value. As such, they have a flat prior belief $\pi(\theta) \propto 1$ about the variable and they resort to consulting J exchangeable human forecasters who share the flat prior about the variable of interest but have access to additional information that will allow them to estimate θ more precisely.

The experts make forecasts in two steps. First, they each form an initial belief about θ before receiving AI advice. Specifically, for all $j = 1, \dots, J$, expert j ’s initial forecast is given by

$$f_j^0 = \theta + s + y + \varepsilon_j^0, \tag{1}$$

where s , y , and ε_j^0 are separate, random errors in their forecast of the target variable θ . We assume that these errors are independent and follow elliptically contoured distributions with mean zero and positive, finite variances σ_s^2, σ_y^2 , and σ_0^2 , respectively. This family includes common distributions such as the multivariate normal distribution, the multivariate t -distribution, and other heavy-tailed distributions such as the multivariate logistic distribution (Cambanis et al. 1981). Subsequently, all experts observe AI advice, which is given by

$$f^a = \theta + s + x, \quad (2)$$

where x is another random error in the AI's forecast of the target variable θ . Similarly to before, the error x is independent and follows an elliptically contoured distribution with mean zero and positive, finite variance σ_x^2 .

The statistical structure of equations (1) and (2) in combination are particularly suitable for modeling algorithm-assisted demand forecasting environments. Such a representation of demand signals is a well-accepted way to reflect the overlapping but nonidentical information sets available to humans and algorithms (Balakrishnan et al. 2025, DosSantos DiSorbo et al. 2025). This model construction can accommodate a wide range of potential variances and correlation structures. For any two human forecasts, the correlation of forecast errors is driven by the variances of their shared components s and y , whereas the correlation of forecast errors between any human forecast and the AI advice is driven only by the variance of their shared component s . The remaining error terms x and ε_j^0 represent residual idiosyncratic errors that are specific to the AI and to expert j , respectively, and show up in neither human-to-human nor human-to-AI forecast error correlations. This model choice also draws a sharp distinction between information contained in AI advice and information contained in advice from humans. Any human advisor in this model would become increasingly redundant as the crowd size J grows large, whereas advice from an AI will continue to hold information that may not be available to any human expert or advisor.

After providing their initial forecast, each expert updates their initial forecast into a final forecast based on the AI advice received. If the experts updated beliefs in an optimal manner, then their final forecast should equal a weighted linear combination of their initial forecast and the AI advice.

PROPOSITION 1. *Expert j 's posterior expectation of θ conditional on their initial forecast and the AI advice, i.e. $f_j^{1*} \stackrel{\text{def}}{=} \mathbb{E}[\theta \mid f^a, f_j^0]$, is*

$$f_j^{1*} = w^* f^a + (1 - w^*) f_j^0, \quad (3)$$

where $w^* = (\sigma_y^2 + \sigma_0^2) / (\sigma_x^2 + \sigma_y^2 + \sigma_0^2) \in (0, 1)$.

Of course, in practice, experts are unlikely to be perfectly rational. To account for any deviations from the normative standard defined in Proposition 1, we generalize the expression in equation (3) to allow for both systematic bias and random noise in the way individual experts update their forecasts after observing the AI advice. First, we assume that on an aggregate level the experts use an *actual* weight on advice $w \stackrel{\text{def}}{=} \alpha w^*$, where the scaling factor $\alpha > 0$ represents the average degree of algorithm aversion in the crowd. Note that we define a crowd’s level of algorithm aversion not in absolute terms, but relative to their optimal utilization of advice, which in our model corresponds to $\alpha = 1$ and a weight on advice of $w = w^*$ (as described in Proposition 1). In our laboratory experiments described in §7, we only observe algorithm aversion (i.e. $\alpha < 1$); however, our analytical results can also be applied to settings where $\alpha > 1$ and experts in the crowd utilize *more* AI advice than they ideally should—a phenomenon known as *algorithm appreciation* (Logg et al. 2019). Second, we include a random, mean-zero error ε_j^1 in each expert j ’s belief about θ after observing f^a , to account for variability in judgmental forecasting accuracy (Moritz et al. 2014). It follows that expert j ’s actual final forecast f_j^1 after receiving AI advice is given by

$$f_j^1 \stackrel{\text{def}}{=} w f^a + (1 - w) f_j^0 + \varepsilon_j^1, \quad (4)$$

where the individual error terms ε_j^1 are independent and follow an elliptically contoured distribution with mean zero and positive, finite variance σ_1^2 conditional on θ, s, x, y , and $\varepsilon_1^0, \dots, \varepsilon_J^0$.

Relaxing the perfectly rational belief updating model of equation (3) to include both the algorithm aversion parameter α and the individual noise term ε_j^1 is necessary to reconcile observed forecasting behavior within a crowd of individuals with our underlying information model. Such behavioral models have a long tradition in the literature and can parsimoniously incorporate both crowd-level biases in how individuals update their beliefs in the face of new information and heterogeneity among the individual experts (Rebholz et al. 2024, Krakowski et al. 2025) while retaining sufficient tractability to derive a prescriptive solution for the DM.

4 Forecast Aggregation and the Borg Effect

The DM’s challenge is to combine all of the information collectively available to the experts together with the AI advice to form the most accurate aggregate estimate of θ to be used in their decision problem. In practice, however, the DM may not observe the AI advice directly, or f^a may be in the form of an unstructured collection of text or information that can only be interpreted and incorporated into a final forecast by the experts. For this reason, we assume that the DM does not have access to f^a and must rely on the individual experts to incorporate the AI advice into an aggregate estimate of θ . This feature is particularly useful in operational contexts where the advice

is generated by proprietary demand forecasting systems or delivered in an unstructured format, which cannot be recorded in a form that the DM can directly aggregate.

One natural approach to aggregation would be to average the experts' individual forecasts after they have observed the AI advice. We define this average final forecast as

$$\bar{f}^1 \stackrel{\text{def}}{=} \frac{1}{J} \sum_{j=1}^J f_j^1. \quad (5)$$

To evaluate the performance of $\bar{f}^1$, we compare it against a hypothetical optimal aggregator known as the *global posterior expectation* (GPE). This represents the aggregator that can observe the AI advice along with the initial forecasts and knows how to combine them in an optimal manner. To make this comparison, we first denote the crowd's average initial forecast as

$$\bar{f}^0 \stackrel{\text{def}}{=} \frac{1}{J} \sum_{j=1}^J f_j^0. \quad (6)$$

PROPOSITION 2. *The crowd's GPE is the posterior expectation of θ conditional on all initial forecasts and the AI advice, i.e. $\theta^* \stackrel{\text{def}}{=} \mathbb{E}[\theta \mid f^a, \{f_j^0\}_{j=1}^J]$, and is given by*

$$\theta^* = \beta w^* f^a + (1 - \beta w^*) \bar{f}^0, \quad (7)$$

where

$$\beta = \frac{\frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2}}{\frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}} \in (0, 1). \quad (8)$$

The GPE specified in equation (7) represents the way in which the DM would like to use the collective information available to the crowd to estimate θ . This ideal, however, differs structurally from the simple average of the experts' final predictions. To understand how, observe that while the descriptive model in equation (4) implies that $\bar{f}^1 = \alpha w^* f^a + (1 - \alpha w^*) \bar{f}^0 + \sum_{j=1}^J \varepsilon_j^1 / J$, which converges to $\alpha w^* f^a + (1 - \alpha w^*) \bar{f}^0$ as the crowd size J increases, the GPE characterized in Proposition 2 is of the form $\theta^* = \beta w^* f^a + (1 - \beta w^*) \bar{f}^0$. A comparison of these two expressions reveals a challenge for the efficiency of $\bar{f}^1$ as an estimator of θ . Even if all of the experts in the crowd optimally incorporate the AI advice into their final forecast (meaning $\alpha = 1$), the process of averaging will push the crowd to collectively place more weight on f^a relative to the average initial forecast $\bar{f}^0$ than it should. This ensuing over-reliance on AI advice, captured by the expression $\beta < 1$, is called the borg effect (Fügener et al. 2021). The closer β is to 0, the stronger the borg effect, with the AI advice receiving $1/\beta$ times more weight in the average final forecast than it would in the GPE. In contrast, as β approaches 1, the borg effect vanishes and the AI advice receives the same weight in the average final forecast as it would in the GPE.

To illustrate the borg effect and its implications for the accuracy of the crowd, consider a specific numerical example.

EXAMPLE 1. Consider a crowd of $J = 9$ perfectly rational demand forecasters who are characterized by the parameters $\alpha = 1$ and $\sigma_1^2 \approx 0$. Furthermore, assume that they are operating in a demand environment in which $\sigma_s^2 = 100$, $\sigma_x^2 = 100$, $\sigma_y^2 \approx 0$, and $\sigma_0^2 = 100$. In this setting, each initial forecast $f_j^0 = \theta + s + y + \varepsilon_j^0$ has the same expected squared forecast error as the AI advice $f^a = \theta + s + x$, and as detailed in equation (3), each expert should give equal weights $w^* = 0.5$ to the AI advice and $1 - w^* = 0.5$ to their own initial forecast when updating their belief, resulting in $f_j^{1*} = 0.5f^a + 0.5f_j^0$. However, in comparison to the experts' collective set of information, the AI advice has much less predictive value. As a result, the GPE would assign the AI advice much less weight relative to $\bar{f}^0$, with equation (8) prescribing $\theta^* = 0.1f^a + 0.9\bar{f}^0$, corresponding to a borg effect of $\beta = 0.2$. Stated differently, the AI advice will receive $1/\beta = 5$ times more weight in the average final forecast than it ideally should. Notably, this misweighting does not emerge because of any irrational behavior on the part of the experts. The AI advice remains useful in this setting and each of the experts is maximizing their individual accuracy by giving it a weight of 0.5. However, the fact that this useful advice is available to all experts, combined with the mechanical process of averaging individual forecasts, ends up undermining the accuracy of the average final forecast because it puts too much stock in the opinion of the AI and too little stock in the collective information held by the human experts.

Example 1 illustrates the potential downsides of AI advice, even when it is helpful at an individual level. Here, each expert's individual initial forecast f_j^0 has an expected squared error of 200, whereas their individual final forecast f_j^1 has a reduced expected squared error of 150. However, the expected squared error of the average initial forecast $\bar{f}^0$ is 111.1, which increases to an expected squared error of 127.8 for the average final forecast $\bar{f}^1$ after the experts observe the AI advice (see §EC.2 of the electronic companion for details). While the individual experts' accuracy improves as a result of the AI advice, the crowd's accuracy worsens, calling into question whether AI advice is a helpful influence from the perspective of the DM.

To quantify the relative improvement in forecast accuracy between the crowd's average initial and average final forecasts, we define the expected squared forecast errors $\mathcal{E}(\bar{f}^0) \stackrel{\text{def}}{=} \mathbb{E}[(\bar{f}^0 - \theta)^2 | \theta]$, and $\mathcal{E}(\bar{f}^1) \stackrel{\text{def}}{=} \mathbb{E}[(\bar{f}^1 - \theta)^2 | \theta]$, and measure the *aggregate benefit of AI advice* $\mathcal{E}(\bar{f}^0) - \mathcal{E}(\bar{f}^1)$. If AI advice increases the accuracy of the crowd, then $\mathcal{E}(\bar{f}^0) > \mathcal{E}(\bar{f}^1)$ and the aggregate benefit of AI advice will be positive, whereas if AI advice decreases the accuracy of the crowd, then $\mathcal{E}(\bar{f}^0) < \mathcal{E}(\bar{f}^1)$ and the aggregate benefit of AI advice will be negative. When the crowd is large, we can characterize this in terms of the borg effect β and the experts' degree of algorithm aversion α .

PROPOSITION 3. *As the crowd size J increases, the aggregate benefit of AI advice converges to*

$$\lim_{J \rightarrow \infty} \mathcal{E}(\bar{f}^0) - \mathcal{E}(\bar{f}^1) = (2\beta - \alpha)\alpha w^{*2}(\sigma_x^2 + \sigma_y^2).$$

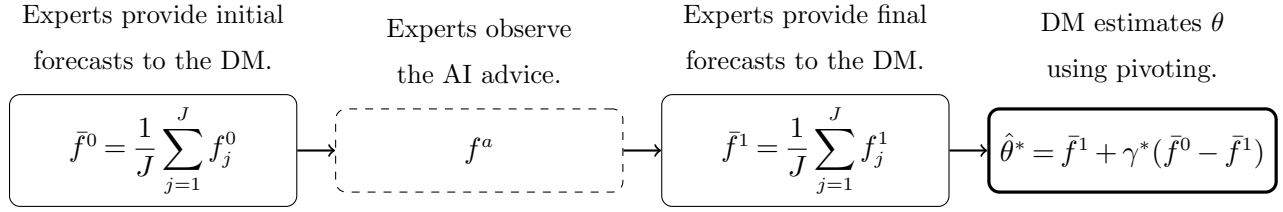
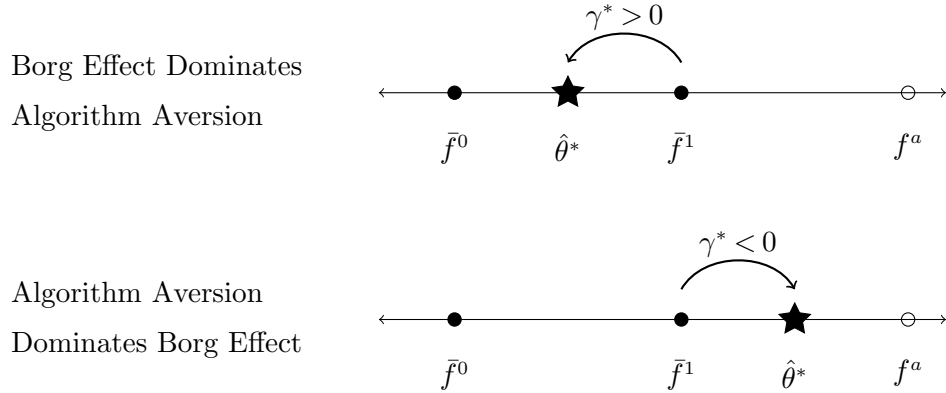
Proposition 3 can help us understand why AI advice was harmful for the crowd in Example 1. Substituting the borg effect $\beta = 0.2$ and the degree of algorithm aversion $\alpha = 1$ in Proposition 3 returns a value of -15 . Note that although this does not specify the aggregate benefit of AI advice exactly because the crowd size $J = 9$ is finite, it provides a close approximation of the difference $\mathcal{E}(\bar{f}^0) - \mathcal{E}(\bar{f}^1) = 111.1 - 127.8 = -16.7$ that appeared in Example 1.

More generally, Proposition 3 reveals two structural characteristics of the aggregate benefit of AI advice for large crowds. First, it implies that the aggregate benefit is positive only when the crowd’s average degree of algorithm aversion satisfies $\alpha \in (0, 2\beta)$. Because the borg effect β is an inherent property of the forecasting environment, this condition translates into a simple threshold: if the crowd places *too much* weight on the AI advice relative to the environment’s borg effect (i.e., $\alpha > 2\beta$), then incorporating AI advice reduces the accuracy of the average final forecast relative to the average initial forecast. In particular, whenever $\beta < 0.5$, a crowd that is “optimally receptive” to advice in the sense that α is close to 1 can be harmed by AI advice. At the other extreme, a crowd with strong algorithm aversion (α close to 0) still benefits from AI advice, although it leaves potential gains on the table by greatly underweighting the advice. Second, the aggregate benefit of AI advice is maximized when the crowd’s average degree of algorithm aversion matches the borg effect, i.e. $\alpha = \beta$. At this point, the crowd underweights the AI advice by exactly the amount needed to counteract the borg effect in the environment, thereby maximizing the accuracy of $\bar{f}^1$.

These results highlight that the value of providing AI advice to the experts depends jointly on the forecasting environment and how the crowd uses the advice. In some settings, the DM may even be better off withholding advice if it is likely to harm the wisdom of the crowd, as demonstrated empirically by Da and Huang (2020). On its face, the DM’s choice appears to be between imperfect utilization of AI advice and not using the advice at all. In the next section, we argue that a better solution is available: even when the crowd’s use of advice is imperfect, the DM can recover much of the lost accuracy by appropriately pivoting the aggregate forecast.

5 Pivoting to Enhance the Crowd’s Forecast Accuracy

As we have seen, when a crowd of experts receives AI advice, neither the average initial forecast nor the average final forecast is guaranteed to be an efficient estimator of θ . At the aggregate level, the final forecast $\bar{f}^1$ can underweight or overweight the AI advice depending on the relative strengths of two opposing forces: algorithm aversion and the borg effect. This motivates an alternative approach that accounts for both rounds of forecasts separately, utilizing the meta-information contained in the adjustment made by the forecasters between the rounds. Because the DM may

Figure 1 Schematic illustration of the process of forecast elicitation and aggregation.**Figure 2** Schematic illustration of the process of estimating the GPE by pivoting when $\gamma^* > 0$ (upper panel) and when $\gamma^* < 0$ (lower panel). $\hat{\theta}^*$ is calculated using only the observable quantities $\bar{f}^1$ and $\bar{f}^0$, which are denoted by solid circles. f^a , which may be unobservable to the DM, is denoted by a hollow circle.

not observe the AI advice in terms of a specific numeric prediction or may not be privy to the advice at all (see §4), the DM's goal is therefore to recover the GPE using only the experts' initial and final forecasts. The next proposition shows how this can be accomplished by *optimally pivoting* from $\bar{f}^1$, using the forecast adjustment $\bar{f}^0 - \bar{f}^1$.

PROPOSITION 4. *The posterior expectation of θ conditional on all initial forecasts and final forecasts, $\hat{\theta}^* \stackrel{\text{def}}{=} \mathbb{E}[\theta \mid \{f_j^0, f_j^1\}_{j=1}^J]$, is given by*

$$\hat{\theta}^* = \bar{f}^1 + \gamma^*(\bar{f}^0 - \bar{f}^1), \quad (9)$$

where

$$\gamma^* = 1 - \frac{\alpha \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right) \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}{\alpha^2 \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right)^2 \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) + \frac{\sigma_1^2}{J}} \in (-\infty, 1). \quad (10)$$

Figure 1 describes in four steps the process by which the DM can generate an efficient estimate of θ by collecting initial and final forecast from experts, and Proposition 4 provides details about the fourth step in particular. Specifically, equation (9) shows that the optimal aggregator of the experts' initial and final forecasts takes the form of pivoting, and the process to recover an efficient estimate θ is governed by the scalar quantity γ^* , which we call the *optimal pivoting factor*.

To develop an intuition for the expression in equation (9) as a solution to the aggregation problem, observe that $\hat{\theta}^*$ begins with the crowd's average final forecast $\bar{f}^1$ and makes an adjustment based on the sign and magnitude of both γ^* and $\bar{f}^0 - \bar{f}^1$. Depending on the sign of γ^* , the pivoting process moves $\bar{f}^1$ either *toward* $\bar{f}^0$ or *away from* $\bar{f}^0$, as illustrated in Figure 2. The upper panel of Figure 2 illustrates a scenario in which the borg effect dominates algorithm aversion, resulting in an average final forecast $\bar{f}^1$ which has moved farther toward f^a than it should to produce the GPE. As a result, the DM should adjust $\bar{f}^1$ back toward $\bar{f}^0$ by a positive pivoting factor $\gamma^* > 0$. In contrast, the lower panel of Figure 2 illustrates a scenario in which algorithm aversion dominates the borg effect, resulting in an average final forecast which has not sufficiently incorporated f^a to arrive at the GPE. Consequently, the DM should adjust $\bar{f}^1$ to move further away from $\bar{f}^0$ by applying a negative pivoting factor $\gamma^* < 0$.

This prescriptive adjustment is structurally analogous to the minimal pivoting procedure proposed by Palley and Soll (2019) but differs methodologically in two key respects. First, the borg effect that arises from the useful but shared AI advice available to the crowd is identified temporally rather than cross-sectionally, emerging as a function of the aggregate shift between the initial and final forecasts. While the additional step of *ex ante* forecast elicitation may come at an incremental cost, the average initial forecast $\bar{f}^0$ plays a crucial role in the pivoting process by providing an appropriate anchor to the average final forecast $\bar{f}^1$. If the DM were to observe only the experts' average final forecast $\bar{f}^1$ without knowing their initial opinion as captured by $\bar{f}^0$, they would not have a way to correct the inefficiency of $\bar{f}^1$ because they would not know which direction to adjust it in (without observing f^a itself). By collecting expert forecasts twice, the DM can observe how the crowd as a whole updates its belief after observing the AI advice, captured by the difference $\bar{f}^0 - \bar{f}^1$. By using the sign and magnitude of this difference together with the optimal pivoting factor γ^* , the DM can recover the most efficient estimate of the variable of interest. Second, the optimal pivoting factor is calibrated to simultaneously correct for both the borg effect and the crowd-level behavioral patterns in how the experts utilize the AI advice, thus incorporating the possibility of the crowd of forecasters being irrational on average.

Importantly, $\hat{\theta}^*$ is a consistent estimator of the GPE as the crowd size J grows large, and the *asymptotically optimal pivoting factor* $\tilde{\gamma} \stackrel{\text{def}}{=} \lim_{J \rightarrow \infty} \gamma^*$ can be expressed as a simple function of the degree of algorithm aversion and the borg effect in the aggregate forecast, as detailed below.

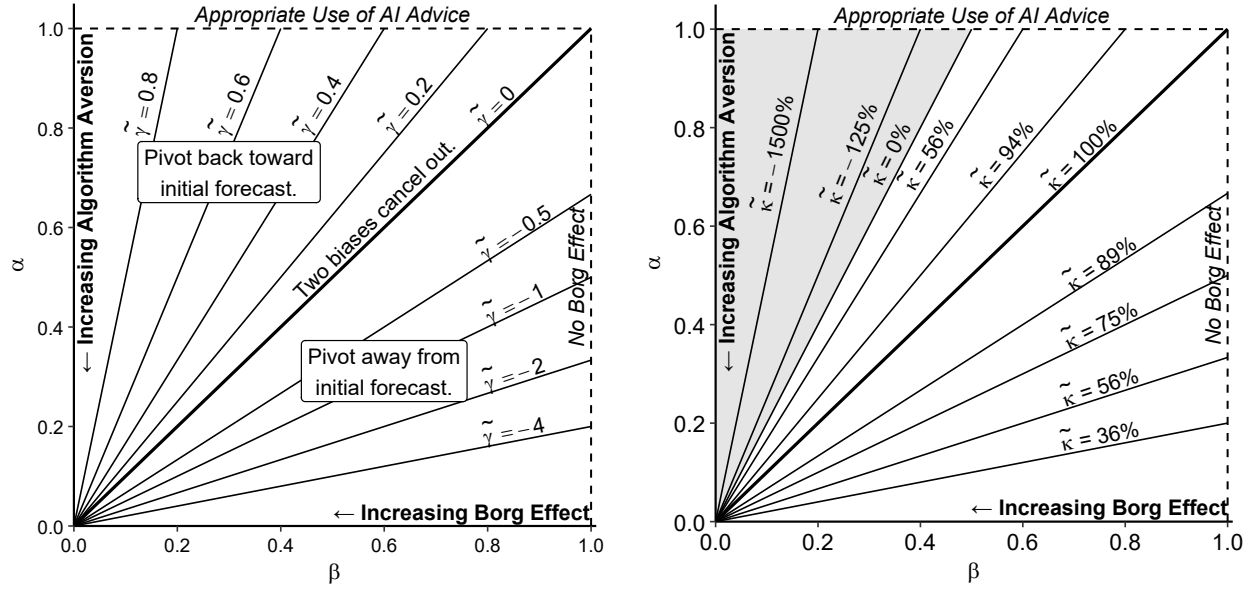
PROPOSITION 5. *As the crowd size increases, $\lim_{J \rightarrow \infty} \hat{\theta}^* = \theta^*$ and the asymptotically optimal pivoting factor is given by*

$$\tilde{\gamma} = 1 - \frac{\beta}{\alpha}. \quad (11)$$

This result indicates that, in theory, the GPE can be recovered from the experts' forecasts by simply increasing the number of forecasters and then applying the optimal adjustment to the crowd's average final forecast using a pivoting process. Further, equation (11) provides a clean interpretation of the asymptotically optimal pivoting factor $\tilde{\gamma}$ as an adjustment that counteracts the joint influence of the borg effect and algorithm aversion on $\bar{f}^1$. At one extreme, there might be no borg effect and no algorithm aversion in the forecasting environment, corresponding to $\beta = 1$ and $\alpha = 1$, respectively. In this case, equation (11) indicates that $\tilde{\gamma} = 0$ and the DM will use $\bar{f}^1$ as their ultimate forecast of θ without the need for any further adjustment. However, when the borg effect is substantial and/or the DM relies on an algorithm-averse crowd, the forecasting environment will be characterized by β and/or α that are less than 1. For these more-typical environments, the extent to which β is less than 1 dictates how much excess weight the estimate $\bar{f}^1$ places on the AI advice as a result of the information structure and the mechanical process of averaging forecasts, while the extent to which α is less than 1 dictates how much excess weight the estimate $\bar{f}^1$ places on the crowd's initial information as a result of the experts' reluctance to utilize the opinion of the algorithm. The size and direction of the recalibration of $\bar{f}^1$ via optimal pivoting depends on the net effect of these two simultaneous biases, which we illustrate using the following example.

EXAMPLE 2. Consider three distinct information environments A , B , and C , which are characterized by $(\alpha_A, \beta_A) = (0.5, 0.25)$, $(\alpha_B, \beta_B) = (0.5, 0.75)$, and $(\alpha_C, \beta_C) = (0.5, 0.5)$, respectively. In all three environments, $\alpha = 0.5$ indicates that the experts are giving the AI advice only half of the weight that they should in order to maximize the accuracy of their own individual final forecasts. Each of the three environments can be visualized within Figure 3a. Information environment A lies in the $\alpha > \beta$ region (upper left) with $\tilde{\gamma}_A > 0$, generating a prescription to pivot $\bar{f}^1$ back toward $\bar{f}^0$. If each expert used the AI advice appropriately to update their individual forecast in this environment, f^a would receive $1/\beta_A = 4$ times the weight in the average final forecast than it should from the DM's perspective. The net effect for a large crowd will be that $\bar{f}^1$ relies too heavily on the AI advice and not enough on the experts' average initial forecast. This inefficiency can be corrected by moving from $\bar{f}^1$ back toward $\bar{f}^0$ using a pivoting factor of $\tilde{\gamma}_A = 1 - \frac{0.25}{0.5} = 1 - 0.5 = 0.5$. In contrast, note that Environment B lies in the $\alpha < \beta$ region (lower right) with $\tilde{\gamma}_B < 0$, generating a prescription to pivot $\bar{f}^1$ away from $\bar{f}^0$. If each expert used the AI advice appropriately to update their individual forecast in this environment, f^a would receive $1/\beta_B = 1.33$ times the weight in the average final forecast than it should from the DM's perspective. The net effect for a large crowd will be that $\bar{f}^1$ relies too heavily on the experts' average initial forecast and not enough on the AI advice. This inefficiency can be corrected by moving from $\bar{f}^1$ away from $\bar{f}^0$ using a pivoting factor of $\tilde{\gamma}_B = 1 - \frac{0.75}{0.5} = 1 - 1.5 = -0.5$. Finally, note that Environment C lies on the identity line $\alpha = \beta$

Figure 3 The optimal pivoting factor and the aggregate AI usage efficiency as a function of the borg effect β and the experts' algorithm aversion α .



(a) Contour plot of the asymptotically optimal pivoting factor $\tilde{\gamma}$.

(b) Contour plot of asymptotic aggregate AI usage efficiency $\tilde{\kappa}$. The shaded region indicates $\tilde{\kappa} < 0\%$.

with $\tilde{\gamma}_C = 0$, and no pivoting is needed. If each expert used the AI advice appropriately to update their individual forecast in this environment, f^a would receive $1/\beta_C = 2$ times the weight in the average final forecast than it should from the DM's perspective. The net effect for a large crowd will be that $\bar{f}^1$ uses the AI advice optimally at an aggregate level, corresponding to an inconsequential pivoting factor of $\tilde{\gamma}_C = 1 - \frac{0.5}{0.5} = 1 - 1 = 0$.

Example 2 provides an overview of three different types of optimal pivoting recommendations that may arise, depending on the parameters α and β that characterize the information environment. Next, to quantify the relative improvement in forecasting accuracy between the crowd's average final forecast and the DM's pivoted forecast, we can define the expected squared forecast error of $\hat{\theta}^*$ as $\mathcal{E}(\hat{\theta}^*) \stackrel{\text{def}}{=} \mathbb{E}[(\hat{\theta}^* - \theta)^2 | \theta]$ and measure the *benefit of pivoting* with $\mathcal{E}(\bar{f}^1) - \mathcal{E}(\hat{\theta}^*)$. If pivoting the average final forecast increases the accuracy of the DM's estimate, then $\mathcal{E}(\bar{f}^1) > \mathcal{E}(\hat{\theta}^*)$ and this benefit is positive. When the crowd is large, we can characterize the benefit of pivoting in terms of the borg effect β and the experts' degree of algorithm aversion α .

PROPOSITION 6. *As the crowd size J grows large, the benefit of pivoting converges to $\lim_{J \rightarrow \infty} \mathcal{E}(\bar{f}^1) - \mathcal{E}(\hat{\theta}^*) = (\alpha - \beta)^2 w^{*2} (\sigma_x^2 + \sigma_y^2)$.*

Proposition 6 indicates that, if the average final forecast is pivoted appropriately, the benefit of pivoting is non-negative and proportional to the squared difference $(\alpha - \beta)^2$. As noted previously,

if $\alpha = \beta$, the pivoted estimate reverts to $\hat{\theta}^* = \bar{f}^1$ and the benefit of pivoting is zero—indicating that there is no need to adjust the crowd’s average final forecast. If $\alpha \neq \beta$, the average weight given to the AI advice in $\bar{f}^1$ will be αw^* while the average weight given to the AI advice in θ^* will be βw^* and the distance between α and β will determine how far $\bar{f}^1$ is from θ^* . Consequently, larger the distance between α and β , greater the need to adjust the crowd’s average final forecast to recover an efficient estimate of θ , and higher the potential benefit of pivoting for the DM.

Together, Propositions 3 and 6 also allow us to characterize the potential aggregate benefit of AI advice for the crowd when the DM utilizes the optimal pivoting procedure. Specifically, as the crowd size grows large,

$$\lim_{J \rightarrow \infty} \mathcal{E}(\bar{f}^0) - \mathcal{E}(\hat{\theta}^*) = \lim_{J \rightarrow \infty} \underbrace{\mathcal{E}(\bar{f}^0) - \mathcal{E}(\bar{f}^1)}_{\text{aggregate benefit of AI advice}} + \lim_{J \rightarrow \infty} \underbrace{\mathcal{E}(\bar{f}^1) - \mathcal{E}(\hat{\theta}^*)}_{\text{benefit of pivoting}} = \beta^2 w^{*2} (\sigma_x^2 + \sigma_y^2) > 0. \quad (12)$$

Equation (12) makes clear that, in contrast to the equivocal aggregate benefit of AI advice to the crowd’s unadjusted average final forecast, the DM’s optimally pivoted estimate $\hat{\theta}^*$ will always be more accurate than the crowd’s average initial forecast. Furthermore, the size of this potential accuracy boost depends on the square of the borg effect parameter β , the optimal individual weight on advice w^* , and the total error variance of the AI advice and the collective human information.

Finally, we define the *aggregate AI usage efficiency*

$$\kappa \stackrel{\text{def}}{=} \frac{\mathcal{E}(\bar{f}^0) - \mathcal{E}(\bar{f}^1)}{\mathcal{E}(\bar{f}^0) - \mathcal{E}(\hat{\theta}^*)}, \quad (13)$$

which we will express as a percentage. For a given forecasting problem, κ measures from the DM’s perspective how efficiently the crowd utilizes the AI advice to improve upon the average initial forecast relative to its potential as represented by the GPE. For example, a forecasting environment with $\kappa = 50\%$ implies that, without a prescriptive pivoting adjustment, the crowd will only capture half of the potential improvement in accuracy that is possible from the AI. On the other hand, a forecasting environment with $\kappa = -50\%$ implies that, without a prescriptive pivoting adjustment, the AI advice will actively harm the crowd’s average forecast accuracy. For large crowds, we can define the asymptotic aggregate AI usage efficiency $\tilde{\kappa} \stackrel{\text{def}}{=} \lim_{J \rightarrow \infty} \kappa$ and it follows directly from Propositions 3 and 6 that we can express $\tilde{\kappa} = (2\beta - \alpha)\alpha/\beta^2 = 1 - \tilde{\gamma}^2/(1 - \tilde{\gamma})^2 \leq 1$.

As indicated in Figure 3b, the estimate $\bar{f}^1$ elicited from a large crowd is only fully efficient, i.e. $\tilde{\kappa} = 100\%$, when $\alpha = \beta$. When α and β do not differ substantially, aggregate AI usage efficiency remains high. However, when there is a meaningful divergence between α and β , $\tilde{\kappa}$ can drop precipitously and, in the extreme, become negative. As highlighted in the shaded upper-left region of Figure 3b, AI advice can harm the accuracy of the unadjusted crowd if $2\beta < \alpha$. More broadly, Figure 3 highlights a key structural relationship between the efficiency of the crowd and the optimal

pivoting procedure—the prescribed pivoting factor identifies the degree of inefficiency in the crowd’s AI-assisted forecast, and corrects it only insofar as an adjustment to $\bar{f}^1$ is indicated.

In sum, our stylized model of information aggregation indicates that the aggregate benefit of AI advice for a crowd of experts depends crucially on the parameters of the forecasting environment and the way in which the experts utilize the advice. We propose a prescriptive pivoting procedure that a DM can use to efficiently combine the unique information held by the human experts with the additional information contained in the AI advice and ensure that providing AI advice to the crowd will result in an unequivocal increase in the accuracy of the DM’s ultimate estimate of the variable of interest. In the next section, we discuss how the DM can implement the pivoting method using historical forecasting data to calculate an appropriate pivoting factor in their unique context.

6 Data-Driven Pivoting: Estimation and Regularization

Although the optimal pivoting method offers the potential to help the DM recover the most efficient estimate of θ given all information observed by the crowd, doing so perfectly requires knowledge of the optimal pivoting factor γ^* for the specific crowd and forecasting problem under consideration. In practice, such precision may be unrealistic because γ^* depends on the way in which the experts in the crowd utilize the AI advice and a number of parameters of the forecasting environment. Instead, practical implementation of the pivoting method requires the DM to leverage any prior information about the crowd to determine a specific pivoting factor γ to be applied to the crowd’s average initial and average final forecasts. To do so, the DM may rely on historical predictions and corresponding demand realizations to empirically learn a suitable value of γ .

To derive a directly computable, data-driven pivoting factor, consider a setting in which the DM has access to crowd forecasts and realized outcomes from N comparable historical forecasting problems. Of note, the validity of our data-driven pivoting estimator relies on the stability of the forecasting environment over time rather than the stationarity of θ itself. For example, the COO of the semiconductor manufacturing firm discussed previously may observe that demand for various chips fluctuates significantly over time, but the broader statistical relationships between the accuracy of the experts, the accuracy of the AI advice, and behavioral patterns in the way the experts use that advice at an aggregate level remain stable. In this setting, the COO can naturally build a history of expert forecasts for near-term demand in each period both before and after receiving AI advice, along with the subsequent, actual demand realization.

For each historical forecasting problem $i \in \{1, \dots, N\}$, we denote the crowd’s average initial forecast, the crowd’s average final forecast, and the realization of θ by $\bar{f}_i^0$, $\bar{f}_i^1$, and θ_i , respectively. For notational convenience, define $X_i \stackrel{\text{def}}{=} \bar{f}_i^1 - \bar{f}_i^0$, $Y_i \stackrel{\text{def}}{=} \bar{f}_i^1 - \theta_i$. Intuitively, X_i measures the size of the crowd’s adjustment after observing the AI advice, while Y_i captures the error in the average final

forecast of the crowd. The DM’s objective is to learn, from these historical pairs, how aggressively to “undo” (or amplify) the crowd’s revision in order to improve accuracy out of sample. Given this data, we define the *empirical pivoting factor* $\hat{\gamma}$ to be the counterfactual pivoting factor that would have minimized the DM’s sum of squared forecast errors over the N historical problems,

$$\hat{\gamma} \in \operatorname{argmin}_{\gamma \leq 1} \sum_{i=1}^N (Y_i - \gamma X_i)^2. \quad (14)$$

Conveniently, the empirical pivoting factor can be calculated as an explicit function of historical average initial forecasts, average final forecasts, and realized values of the uncertain variable of interest, as detailed in the following proposition.

PROPOSITION 7. *Assume that the N historical tasks are i.i.d. draws from the same environment, and the non-degeneracy condition $\sum_{i=1}^N X_i^2 > 0$ holds. The DM’s empirical pivoting factor $\hat{\gamma}$ is given by*

$$\hat{\gamma} = \min \left\{ \sum_{i=1}^N X_i Y_i / \sum_{i=1}^N X_i^2, 1 \right\}. \quad (15)$$

Proposition 7 presents a solution to the constrained minimization problem laid out in equation (14). The i.i.d. condition corresponds to a stable forecasting environment across the historical tasks used for training the empirical model, and the non-degeneracy condition $\sum_{i=1}^N X_i^2 > 0$ specifies that an in-sample risk minimizing estimate of γ^* cannot be calculated if the crowd has never updated its forecast upon receiving AI advice within the DM’s observed sample of historical forecasting tasks. Importantly, it provides a single, univariate pivoting factor $\hat{\gamma}$ that corrects for the joint influence of the borg effect and the degree of algorithm aversion on the accuracy of the crowd forecasts without having to disentangle and separately estimate the two effects. Moreover, when the empirical pivoting factor $\hat{\gamma}$ is less than one, it also matches the estimated slope coefficient from the following ordinary least squares regression without an intercept: $\bar{f}_i^1 - \theta_i = \gamma(\bar{f}_i^1 - \bar{f}_i^0) + \xi_i$. When J is large, we can also use $\hat{\gamma}$ to estimate an empirical proxy of the aggregate AI usage efficiency defined in equation (13) according to

$$\hat{\kappa} \stackrel{\text{def}}{=} 1 - \hat{\gamma}^2 / (1 - \hat{\gamma})^2. \quad (16)$$

In §EC.3 of the electronic companion, we demonstrate the asymptotic consistency of the empirical pivoting factor $\hat{\gamma}$ and show that it converges to the true parameter γ^* at a rate of $\mathcal{O}(N^{-1/2})$. We also construct worst-case high-probability bounds on the finite-sample performance of the empirical pivoting method and demonstrate that the loss in predictive performance due to estimation decreases at a relatively faster rate of $\mathcal{O}(N^{-1})$.

6.1 A Method to Regularize the Estimated Pivoting Factor

When the crowd’s adjustments after seeing AI advice are generally small, the denominator $\sum_{i=1}^N X_i^2$ in equation (15) can be small and the unregularized estimator becomes sensitive to noise, which can lead to aggressive pivoting recommendations that are difficult to justify from limited data. Moreover, when the true benefit of pivoting is modest, an unregularized estimate may recommend a nonzero adjustment simply because it fits idiosyncrasies of the historical sample. This motivates a conservative variant that would default to *null pivoting*—a recommendation to not adjust the average crowd forecast unless the historical data provides strong evidence that pivoting is beneficial.

To stabilize the empirical estimation process when the size of the historical dataset is small or when the adjustment signal is weak, we propose an ℓ_1 -penalized (LASSO) estimate of the pivoting factor (Tibshirani 1996). Define the *regularized pivoting factor* $\hat{\gamma}_\lambda$ as the solution to the penalized squared error minimization problem

$$\hat{\gamma}_\lambda \stackrel{\text{def}}{=} \underset{\gamma \in (-\infty, 1]}{\operatorname{argmin}} \sum_{i=1}^N (Y_i - \gamma X_i)^2 + \lambda |\gamma|, \quad (17)$$

where $\lambda \geq 0$ is a regularization parameter that controls the degree of shrinkage. Subsequently, the regularized pivoting factor produces a conservative *regularized pivoted estimate* of θ , given by

$$\hat{\theta}_\lambda \stackrel{\text{def}}{=} \bar{f}^1 + \hat{\gamma}_\lambda (\bar{f}^0 - \bar{f}^1). \quad (18)$$

The next proposition provides a closed-form expression for $\hat{\gamma}_\lambda$, and identifies the condition when null pivoting, i.e. $\hat{\gamma}_\lambda = 0$, is recommended. For notational convenience, define $S \stackrel{\text{def}}{=} \sum_{i=1}^N X_i Y_i$, $Q \stackrel{\text{def}}{=} \sum_{i=1}^N X_i^2$.

PROPOSITION 8. *Assuming $Q > 0$, the unique, closed-form solution to the penalized minimization problem in (17) is*

$$\hat{\gamma}_\lambda = \min \left\{ \frac{\operatorname{soft}(S, \lambda/2)}{Q}, 1 \right\}, \quad (19)$$

where the *soft-thresholding operator* $\operatorname{soft}(s, a) \stackrel{\text{def}}{=} \operatorname{sgn}(s)(|s| - a)^+$, $\operatorname{sgn}(z) = z/|z|$ (for $z \neq 0$), and $(z)^+ = \max\{z, 0\}$. Further, $\hat{\gamma}_\lambda = 0$ occurs whenever $|S| \leq \lambda/2$.

Proposition 8 makes clear why an ℓ_1 -penalty is well-suited for practical pivoting. The regularized estimator is driven by the empirical covariance S between the crowds’ adjustments and the errors in the average final forecasts. When this signal is weak relative to the penalty level λ , the procedure sets $\hat{\gamma}_\lambda = 0$ (null pivoting), so the DM defaults to the conservative baseline $\bar{f}^1$ rather than extrapolating based on noise in a small or uninformative historical sample. When the signal is strong, $\hat{\gamma}_\lambda$ instead behaves like a shrunk version of the unregularized slope, reflecting the familiar bias–variance trade-off: regularization introduces some shrinkage bias in exchange for improved

finite-sample stability and protection against extreme pivoting recommendations. The specific value of the regularization parameter λ controls how conservative the estimator $\hat{\gamma}_\lambda$ is, where larger values of λ produce smaller, more stable coefficients. In practice, cross-validation methods are commonly used to select a λ that stabilizes the finite-sample estimation without taking on disproportionate risk, which we discuss and implement in §7.1.5 of the paper. In §EC.3 of the electronic companion, we characterize the irreducible bias induced by regularization as a direct cost of controlling against the variance due to unstable estimation, and show that the size of the bias grows at a rate of $\mathcal{O}(\lambda)$. We also construct worst-case high-probability bounds on finite-sample performance of the regularized empirical pivoting method, and demonstrate that the loss in predictive performance due to regularization increases at a relatively faster rate of $\mathcal{O}(\lambda^2)$.

7 An Experimental Test of the Pivoting Method

To test the performance of the data-driven pivoting method discussed in the previous section, we designed and carried out a controlled experiment in which participants made quantitative predictions both before and after receiving AI advice. Specifically, we repurposed a task from the advice-taking literature which asked participants to predict the ages of people pictured in a photograph (see, e.g., Soll et al. (2022)). Participants twice provided estimates about the person’s age—once before and once after seeing the advice. AI assistance was generated using OpenAI’s GPT-4 model and provided to participants in the form of a numerical prediction along with a text-based explanation supporting that prediction. This task provides us with a forecasting context that is well-suited to test the mechanism and prescription proposed by our theory. It recreates the proposed structure of judgmental adjustments in AI-assisted forecasting tasks in a setting that is experimentally accessible: participants form their initial belief, the AI advice provides a common but informative opinion, and participants update their judgment accordingly. In addition, our chosen task utilizes a forecasting skill that online participants have developed naturally through their everyday lives and the experimental design allows us to artificially control the effectiveness of the AI advice while holding all else about the forecasting environment fixed, as we discuss next.

The experiment consisted of three separate studies, carried out in phases, each of which treated participants with AI advice of different accuracies. By doing so, we were able to explore a variety of forecasting environments with model parameters that would prescribe different directions and magnitudes of pivoting for the aggregator while keeping the forecasting task and context consistent across treatments (Walzner et al. 2022). To generate such variation in forecasting environments, we modified the standalone accuracy of AI advice as follows: In the *Baseline* condition, the AI advice was drawn directly from GPT-4 and provided to the participants. In the *Worse Advice* treatment, we increased the error magnitudes in the AI advice from the Baseline condition in a controlled

manner, whereas in the *Better Advice* treatment, we reduced the error magnitudes in the AI advice from the Baseline condition in a controlled manner. Preregistration documents, experiment materials, recorded data, and code to implement the regularized pivoting method and analyze the results of the online experiment consisting of three treatment conditions are provided at the following link: https://osf.io/pjha6/overview?view_only=e915a166c08c44dc8a8638b0aa499b76.

7.1 Experimental Design and Analysis

7.1.1 Materials. The images used for the age estimation task in the experiment were drawn from the *One to One Hundred Years Project* by portrait photographer Edouard Janssens. The entire collection of photographs can be accessed at <https://edouardjanssens.com/art/1-to-100-years/index.html>. This collection included photographs of 100 different males and 100 different females, representing one person for each gender for every age from 1 year to 100 years old, giving us a total of 200 images that we could display to participants. The photographs were taken in a consistent style with relatively homogeneous facial expressions. For each of the 200 tasks, the true age of the person in the photograph (which was unknown to participants) served as the target variable being estimated.

7.1.2 Participants. We recruited participants from the crowdsourcing platform Prolific. The three studies were carried out separately and in sequence over time, beginning with Baseline, followed by Worse Advice, and concluding with Better Advice. Every participant was assigned to a single treatment and was asked to provide age estimates for 10 faces that were independently sampled without replacement from the 200 photographs and presented in a random order. Our preregistered data collection plans called for targets of 1100 participants in the Baseline condition and 1050 participants each in the Worse Advice and Better Advice conditions, with 10 age estimate response-pairs (before and after observing the AI advice) elicited from each participant in each treatments. After removing responses from participants who were flagged by the exclusion criteria, we obtained 10,930, 10,470, and 10,480 response-pairs in the Baseline, Worse Advice, and Better Advice treatment conditions, respectively. Mean completion times for the retained participants were 606 seconds, 600 seconds, and 558 seconds, in the three respective treatments.

7.1.3 Procedure. The experiment was programmed in Qualtrics and completed through an online interface. Before starting the main experiment, participants completed two comprehension checks about the payout structure and three practice questions to familiarize themselves with the experimental setup. After completing the comprehension checks about bonus payments and practicing the age estimation task for three photographs, participants moved on to the main study. For each of the 10 tasks in the main study, participants were first presented with the photograph of the person’s face and asked to provide an initial estimate of their age. After submitting this response,

the participant moved on to another screen that displayed the same photograph again along with AI advice consisting of a numeric point estimate of the person’s age and a short paragraph of text in which GPT-4 provided a justification of its estimate. After each stage, participants were prevented from going back to previous screens and modifying their response to a preceding question.

7.1.4 Payments. Each participant was paid \$1 for completing the study. Additionally, participants were incentivized to be as accurate as possible when providing their estimates through separate bonus payments for both the initial and final ages estimates for each of the 10 tasks. Participants could earn a maximum bonus of 10¢ for each age estimate, with 1¢ deducted for every year of difference from the true value and no bonus for estimates that were off by 10 years or more. For example, if the actual age of the person in the photograph was 53 years, then an estimate of 47 years would earn a bonus of 4¢, whereas an estimate of 63 years would earn a bonus of 0¢. The average total bonuses earned by the retained participants in the Baseline condition, Worse Advice condition, and Better Advice condition were \$0.97, \$0.96, and \$1.02, respectively.

7.1.5 Hypotheses. Based on the theoretical model developed in §5, we predicted that the crowd’s pivoted forecasts would provide the most accurate estimates in each of the three forecasting environments. We formalize this according to the following three hypotheses:

HYPOTHESIS 1. Across all three treatments, the crowd’s regularized pivoted forecast $\hat{\theta}_\lambda$ will provide a lower mean squared forecast error than the crowd’s average initial forecast $\bar{f}^0$.

HYPOTHESIS 2. Across all three treatments, the crowd’s regularized pivoted forecast $\hat{\theta}_\lambda$ will provide a lower mean squared forecast error than the AI advice f^a .

HYPOTHESIS 3. Across all three treatments, the crowd’s regularized pivoted forecast $\hat{\theta}_\lambda$ will provide a lower mean squared forecast error than the crowd’s average final forecast $\bar{f}^1$.

To test these hypotheses, we utilize a holdout procedure within each treatment to apply the data-driven pivoting method described in §6 to the average initial and average final forecasts for all 200 photographs. To avoid data leakage, we separately estimate the pivoting factor for each photograph. First, we set one photograph and its average initial and final age estimates aside. Second, over the remaining $N = 199$ holdout tasks, we use expressions (15) and (17) to compute the empirical pivoting factor and the regularized pivoting factor, respectively. For the regularized pivoting factor, we select λ via 10-fold cross-validation using the *glmnet* package in R (Friedman et al. 2025). Finally, we use the estimate of $\hat{\gamma}_\lambda$ to calculate a conservatively pivoted estimate of θ described in equation (18), i.e. $\hat{\theta}_\lambda$, for the task that was set aside in step one.

Figure 4 Parameter estimates of $\hat{\alpha}$, $\hat{\beta}$, $\hat{\kappa}$, and $\hat{\gamma}$ across the three treatment conditions for each of the 200 forecasting tasks in the three treatment conditions.

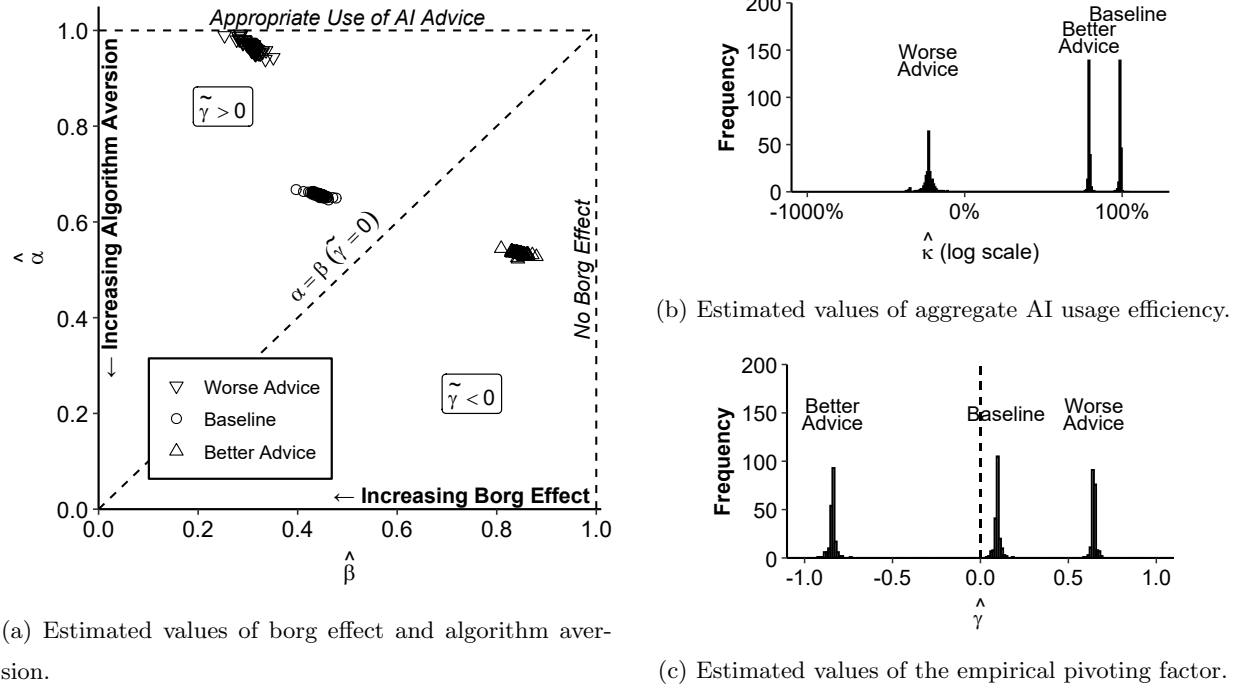


Table 1 Descriptive Statistics of the Estimated Parameters in the Three Information Environments.

Parameter	Worse Advice	Baseline	Better Advice
$\hat{\alpha}$	0.97 ± 0.007	0.66 ± 0.002	0.53 ± 0.002
$\hat{\beta}$	0.31 ± 0.009	0.44 ± 0.008	0.84 ± 0.007
$\hat{\kappa}$	$-233\% \pm 35.2\%$	$99\% \pm 0.5\%$	$79\% \pm 0.5\%$
$\hat{\gamma}$	0.64 ± 0.012	0.10 ± 0.016	-0.84 ± 0.019

Notes: $\hat{\alpha}$, $\hat{\beta}$, $\hat{\kappa}$, and $\hat{\gamma}$ are estimated using equations (EC.4), (EC.3), (16), and (15), respectively. Estimates are reported as Mean $\pm$ Standard Deviation.

7.2 Results

7.2.1 Description of the Experimental Forecasting Environments. First, we check to make sure that the participants in the three treatment groups are not characteristically different (with the exception of the quality of the AI advice provided to them). To do so, we compare the accuracies of the crowd's average initial forecast across treatments for each photograph. We find that the average squared forecast errors of $\bar{f}^0$ are not significantly different ($p = 0.479$ from a parametric analysis-of-variance (ANOVA) test; $p = 0.497$ from a nonparametric Friedman test), with a root mean squared error (RMSE) of 6.54 years in the Baseline treatment, 6.46 years in the Better Advice treatment, and 6.42 years in the Worse Advice treatment.

We begin by exploring the characteristics of the three forecasting environments generated by the three experimental treatments (see Table 1 for a summary). For each of the 200 photographs within

each treatment, we can estimate the degree of algorithm aversion (defined as $\hat{\alpha}$) and the borg effect (defined as $\hat{\beta}$) as discussed in §EC.1.2 of the electronic companion, and can use equation (15) to estimate $\hat{\gamma}$ and equation (16) to estimate $\hat{\kappa}$ similarly for every treatment-task combination. The resulting pairs of $\hat{\alpha}$ and $\hat{\beta}$ estimates are shown in Figure 4a, and histograms of $\hat{\kappa}$ and $\hat{\gamma}$ are shown in Figures 4b and 4c respectively. Of note, $\hat{\alpha}$ and $\hat{\beta}$ can be separately estimated only in cases in which a numeric value of AI advice is observed for all historical forecasts. However, estimation of $\hat{\kappa}$ and $\hat{\gamma}$ is possible even if the AI advice is non-numeric or unknown.

We observe in Figure 4a that the Baseline treatment reflects an information environment in which the two biases are roughly equal in magnitude and cancel each other out. In line with the intuition that little or no adjustment of the crowd’s average final estimate $\bar{f}^1$ is necessary in such settings, we also observe that the distribution of $\hat{\kappa}$ is close to 100% and the distribution of $\hat{\gamma}$ is close to zero in this treatment—as seen in Figures 4b and 4c respectively. In contrast, pivoting may be beneficial in the other two forecasting environments. In the Worse Advice treatment, the forecasters are, on average, giving the AI advice an appropriate weight, but the common advice induces a strong borg effect in the average final forecasts. In fact, the borg effect is severe enough here that the provision of AI advice is expected to end up harming the crowd’s accuracy at an aggregate level and the estimated AI usage efficiencies $\hat{\kappa}$ are negative. As a result, the empirical pivoting factors $\hat{\gamma}$ are positive—indicating that $\bar{f}^1$ needs to be adjusted back toward $\bar{f}^0$. In the Better Advice treatment, the borg effect is relatively weak but the forecasters are placing far less weight on the AI advice than they should, indicating a strong degree of algorithm aversion. The estimated AI usage efficiencies here $\hat{\kappa}$ are positive but less than 100% and the empirical pivoting factors $\hat{\gamma}$ are negative—indicating that the AI advice is not being used efficiently to improve the crowd’s accuracy and that $\bar{f}^1$ needs to be adjusted further away from $\bar{f}^0$.

7.2.2 Evaluating the Accuracy of the Pivoting Method. With this background, we can assess the forecasting accuracy of the regularized pivoted estimates within each of the three treatments and compare them to the benchmark forecasts $\bar{f}^0$, f^a , and $\bar{f}^1$. To do so, we calculate the squared forecasting error for each method for each photograph within each treatment. Results are summarized in Table 2. After calculating these errors, we can formally test our three main hypotheses within each treatment. Two important deviations from our preregistered analysis plans should be noted here. First, the method by which the pivoting factor was estimated changed over the course of our research and the LASSO-regularized pivoting method was only specified for the latest treatment (“Better Advice”). The preregistration for the initial Baseline treatment specified a method for estimating the pivoting factor within a single task that relied on the DM observing the numeric AI advice, and the preregistration for the subsequent Worse Advice treatment specified a

Table 2 Estimated Regularized Pivoting Factors and Relative Performances of the All Forecasts.

	Worse Advice	Baseline	Better Advice
N	200	200	200
Average $\hat{\gamma}_\lambda$	0.64	0.00	-0.83
RMSE of the Average Initial Forecast ($\bar{f}^0$)	6.42	6.54*	6.46***
RMSE of the AI Advice (f^a)	11.34***	7.07***	5.64
RMSE of the Average Final Forecast ($\bar{f}^1$)	6.68*	6.28	5.71
<i>RMSE of the Pivoted Forecast $\hat{\theta}_\lambda$</i>	6.36	6.28	5.54

Notes: RMSE is measured in years. Numbers in bold denote the lowest RMSE within each treatment.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ for a two-tailed paired t -test of means. Statistical significance is assessed by comparing the squared errors of $\hat{\theta}_\lambda$ and the three benchmark forecasts $\bar{f}^0$, f^a , and $\bar{f}^1$.

method for estimating the pivoting factor using a complex regret-minimization method. Second, our planned hypothesis tests for the initial Baseline treatment and subsequent Worse Advice treatment specified using absolute forecast errors. To align with the forecasts as estimates of the expectation of θ , and for consistency of presentation, we test our comparisons using squared errors as specified in the preregistration for the latest treatment (“Better Advice”). Postmortem documents detailing results for the original, preregistered hypothesis tests in the Baseline and Worse Advice treatments are available in the OSF repository with the preregistration documents.

In the Baseline treatment, the regularized pivoting factors are equal to zero, so the pivoted forecasts revert to null pivoting, with $\hat{\theta}_\lambda = \bar{f}^1$ for all 200 photographs, and there are no differences between the two methods. The pivoted forecasts are significantly more accurate than the AI advice itself (paired $t_{199} = -3.68$, $p < 0.001$) but are not significantly more accurate than the average initial forecasts (paired $t_{199} = -1.88$, $p = 0.062$). In the Worse Advice treatment, the pivoted forecasts are significantly more accurate than the AI advice (paired $t_{199} = -6.50$, $p < 0.001$) but are not significantly more accurate than the average initial forecasts (paired $t_{199} = -0.57$, $p = 0.564$) or the average final forecasts (paired $t_{199} = -1.70$, $p < 0.091$). In the Better Advice treatment, the pivoted forecasts are significantly more accurate than the average initial forecasts (paired $t_{199} = -3.41$, $p < 0.001$) but are not significantly more accurate than the AI advice (paired $t_{199} = -1.12$, $p = 0.261$) or the average final forecasts (paired $t_{199} = -1.37$, $p = 0.172$).

To understand these results, we consider the average amounts of adjustment recommended by the regularized pivoting method and the aggregate AI usage efficiencies in each treatment. Observe from Table 2 that in the Baseline condition, as a consequence of the two biases canceling out, $\hat{\gamma}_\lambda$ is reduced to zero for all tasks. In contrast, in both the Better Advice and Worse Advice conditions, the regularized pivoting factor remains noticeably away from zero, underscoring the presence of a net bias in the final consensus forecast—and therefore a need to pivot. In the Worse Advice

condition, the mean adjustment is to pivot back toward $\bar{f}^0$ by a factor of 0.64, whereas in the Better Advice condition the mean adjustment is to pivot away from $\bar{f}^0$ by a factor of 0.83.

Importantly, in all three treatments, regularized pivoting proves to be the most accurate method as measured by RMSE of the forecasts. In the Baseline condition, $\bar{f}^1$ has the lowest RMSE of the three benchmark forecasting methods and, since $\hat{\gamma}_\lambda = 0$ for all tasks, its performance is exactly replicated by $\hat{\theta}_\lambda$. This corroborates the evidence from Figure 4b that the crowd’s AI-assisted forecasts in the Baseline condition were highly efficient to begin with. In the Worse Advice condition, $\bar{f}^0$ has the lowest RMSE of the three benchmark forecasting methods. Here the crowd’s accuracy actually worsens after incorporating AI advice in this treatment, with the RMSE of the average forecasts increasing from 6.42 years to 6.68 years, and this detrimental effect of AI advice is reflected in the negative $\hat{\kappa}$ values in Figure 4b. However, after pivoting back toward the average initial crowd forecast, $\hat{\theta}_\lambda$ delivers an RMSE of 6.36 that is lower than all three of the benchmark methods, finding a sweet spot roughly two-thirds of the way between $\bar{f}^1$ and $\bar{f}^0$. The improvement of $\hat{\theta}_\lambda$ over $\bar{f}^1$ is statistically significant primarily due to the very poor (negative) efficiency of $\bar{f}^1$ in this treatment. In the Better Advice condition, $\bar{f}^a$ has the lowest RMSE of the three benchmark forecasting methods. Although the crowd’s accuracy increases after receiving AI advice, lowering the RMSE of their forecasts from 6.46 years to 5.71 years, the participants display algorithm aversion and, as a result, do not outperform the RMSE of the AI advice itself of 5.64 years. Pivoting away from $\bar{f}^0$ in this case allows the DM to take full advantage of the superior accuracy of the AI advice, even without observing its specific numeric forecast or the verbal reasoning behind it. The regularized pivoted forecasts in this treatment deliver a reduced RMSE of 5.54 years, which is lower than any of the three benchmarks. However, the improvement of $\hat{\theta}_\lambda$ over $\bar{f}^1$ is statistically insignificant in this treatment because the efficiency of $\bar{f}^1$, albeit lower than 100%, was relatively high to begin with. Overall, we find that the pivoting method offers robust performance across three very different treatments and adjusts to the crowd’s final average forecast only insofar as the inefficiency in the forecasting environment requires it.

8 Discussion

As AI capabilities continue to rapidly advance, algorithmic assistance is becoming an essential component of many business problems, including demand forecasting and supply chain planning. While a simple average of individual forecasts obtained from a crowd of experts serves as a well-performing aggregation strategy, there is empirical evidence that shared information can limit the potential accuracy improvements (Palley and Soll 2019). When shared information is in the form of AI advice that is available to all experts, this phenomenon is referred to as the “borg effect”, which can lead to the loss of unique human insight into the problem (Fügener et al. 2021).

Table 3 Schematic Representation of Four Potential Combinations of Borg Effect and Algorithm Aversion and Their Implications for Pivoting.

	Low Algorithm Aversion	High Algorithm Aversion
Low Borg Effect	$\gamma = 0$ (Stick with the final forecast.)	$\gamma < 0$ (Pivot away from the initial forecast.)
High Borg Effect	$\gamma > 0$ (Pivot back toward the initial forecast.)	$\gamma = 0$ (Stick with the final forecast.)

In this paper, we argue that the DM can always benefit from providing assistance to demand forecasters, especially when that information is produced by well-trained algorithms. The pitfalls of providing AI advice to a crowd can be overcome by eliciting *ex ante* predictions before advice is shared with the forecasters, and adjusting the consensus forecast using the pivoting method proposed in §5. Importantly, pivoting is not simply a convex combination of two forecasts; instead it exploits the structure created by the process of providing experts with AI assistance, using the fact that the same crowd produces forecasts both before and after receiving common AI advice to diagnose and correct systematic misweighting of information at the aggregate level. As observed in equation (11), the asymptotically optimal pivoting factor simultaneously depends on two parameters α and β which measure the degree of two sources of misweighting in the aggregate forecast.

Interestingly, the two forces act in opposite directions and are capable of negating each other's effect. Broadly speaking, we can think of the potential forecasting environments as falling into one of four categories based on whether the experts' algorithm aversion is low or high and whether the borg effect is low or high, as represented schematically in Table 3. When the two forces are similar, the crowd's aggregate AI usage is close to fully efficient (as observed from Figure 3b). A key characteristic of the proposed pivoting procedure is that it is able to identify such contexts and prescribe a pivoting factor that is close to zero. Stated differently, in these situations the DM can simply take the crowd's highly efficient final forecast as given. We call this conservative approach *null pivoting*, which acts as a guardrail against harming the AI-assisted crowd when there is no need for any adjustment. However, when the two forces differ substantially, the DM can improve the crowd's forecast by pivoting the average final forecast either away from or back toward the average initial forecast, depending on the relative magnitudes of α and β .

If historical forecasting data is available, the DM can determine a specific pivoting factor using the data-driven method detailed in §6.1, providing a prescriptive adjustment that is well-adapted to the forecasting environment. An attractive feature of the proposed method is that the DM does not need to observe the AI advice or associate it with a specific numerical prediction in order to use this technique. With the advent of generative AI and proprietary machine learning models,

the advice available to experts may itself be unstructured, latent, or unavailable to the DM. Even when the advice itself remains unobserved, it is possible for the DM to determine the *effect* of the advice on individual beliefs by recording forecasts both before and after the advice is received, and estimate a pivoting factor from historical tasks using only $(\bar{f}^0, \bar{f}^1, \theta)$.

Our controlled experiment provides evidence that, indeed, the pivoting method can boost accuracy for real human forecasters, with potential benefits that may vary depending on the specific task, the crowd, and the AI advice. We separately observe forecasting environments for which positive and negative pivoting significantly reduces the crowd’s forecasting error, in the Worse Advice and Better Advice conditions respectively. In both cases, pivoting in the appropriate direction and by the prescribed amount led to more accurate forecasts that match or surpass the crowd’s average initial forecast, average final forecast, and the AI advice itself. We also observe a forecasting environment where borg effect and algorithm aversion are roughly equal in magnitude; in the Baseline condition of our experiment, the regularized pivoting factor is shrunk to zero in every task, and pivoting makes no modification to the crowd’s average final forecast. Viewed differently, the Baseline condition demonstrates that a crowd of AI-assisted experts can potentially arrive at a very good forecast in spite of the limiting presence of both algorithm aversion and borg effect—and as such it is appropriately identified by the regularized pivoting method. This is a somewhat counterintuitive finding in light of recent academic literature focused on overcoming or mitigating inefficient use of algorithmic advice (Dietvorst et al. 2018), raising the possibility that algorithm aversion can be overcome by utilizing a crowd of forecasters.

To conclude, firms implementing AI-assisted demand forecasting should consider how to optimally integrate the complementary strengths of algorithms and humans (Davis et al. 2024). Some existing research (Da and Huang 2020, Fügener et al. 2021) presents an argument in favor of not revealing public information or algorithmic advice to forecasters, even when it might improve their individual accuracy—because the crowd consensus is likely to be affected by shared-information bias which results in a loss of valuable private information. We argue that there exists a better way to utilize potentially beneficial AI advice while simultaneously preserving the value of private information and adjusting for algorithm aversion in experts. In operational terms, the DM’s choice is not limited to “show advice versus withhold advice”: pivoting provides an auditable post-processing lever that preserves informational benefits of AI advice while mitigating its potential to distort crowd wisdom. The theoretical foundations of our method combined with a prescriptive, data-driven mode of application are well-suited for implementation by operations managers who rely on human-AI collaborative teams to generate demand forecasts.

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Electronic Companion to *Enhancing the Wisdom of Crowds in AI-Assisted Judgmental Forecasts*

EC.1 Some Useful Additional Results

EC.1.1 Calculating the Posterior Expectation of θ Using Point Forecasts

Define $\mathbf{g} \stackrel{\text{def}}{=} \{g_1, \dots, g_p\} \in \mathbb{R}^p$ as a vector of length $p \in \mathbb{N}$, containing p point forecasts of θ . Assume a flat prior $\pi(\theta) \propto 1$ about θ . Define $\Sigma \stackrel{\text{def}}{=} \text{Cov}[\mathbf{g} \mid \theta]$ as a symmetric, positive definite covariance matrix of $\mathbf{g}$ having dimension $p \times p$. Further, let $\mathbf{e}$ is a unit vector of length p .

LEMMA EC.1. *If the forecast errors conditional on θ , i.e. $\mathbf{g} - \theta\mathbf{e} \mid \theta$, belong to the family of elliptically contoured distributions with zero mean and positive, finite variance, then the posterior expectation of θ conditional on $\mathbf{g}$ and Σ , i.e. $\mathbb{E}[\theta \mid \mathbf{g}, \Sigma]$, is given by $\boldsymbol{\omega}^{*\top} \mathbf{g}$, where*

$$\boldsymbol{\omega}^* = \frac{\Sigma^{-1} \mathbf{e}}{\mathbf{e}^\top \Sigma^{-1} \mathbf{e}}. \quad (\text{EC.1})$$

Proof: We consider a linear class of estimators of θ where the posterior expectation conditioning on the error covariance structure is a linear combination of the unbiased forecasts. By restricting the mean-zero forecast errors to the family of elliptically contoured distributions, the best estimator is guaranteed to be a linear combination of the individual forecasts, therefore coinciding with the best linear estimator (see Theorem (2.18), pp. 45 of Fang et al. (2018)). Next, we argue that in order to return the unbiased posterior expectation of θ , $\mathbf{e}^\top \boldsymbol{\omega} = 1$ is a necessary condition on $\boldsymbol{\omega}$. Note that $\mathbb{E}[\mathbf{g} \mid \theta] = \theta\mathbf{e}$ (since the individual forecasts are unbiased), and $\mathbb{E}[\boldsymbol{\omega}^\top \mathbf{g} \mid \theta] = \theta$ (since the posterior expectation is unbiased). Therefore, $\theta = \mathbb{E}[\boldsymbol{\omega}^\top \mathbf{g} \mid \theta] = \boldsymbol{\omega}^\top \mathbb{E}[\mathbf{g} \mid \theta] = \theta \boldsymbol{\omega}^\top \mathbf{e}$, and $\mathbf{e}^\top \boldsymbol{\omega} = 1$.

By definition, among all real-valued functions $h : \mathbb{R}^p \rightarrow \mathbb{R}$, the posterior expectation $h(\mathbf{g}) := \mathbb{E}[\theta \mid \mathbf{g}]$ is the function that minimizes the Bayes risk $\mathbb{E}[(h(\mathbf{g}) - \theta)^2]$ over the joint distribution of $\mathbf{g}$. We can argue that conditional on θ , $\text{Cov}[\mathbf{g} - \theta\mathbf{e} \mid \theta] = \text{Cov}[\mathbf{g} \mid \theta] = \Sigma$. Therefore the Bayes risk (a scalar quantity) of a linear combination $\boldsymbol{\omega}^\top \mathbf{g}$ conditional on θ and Σ can be expressed in the quadratic form $\text{Var}[\boldsymbol{\omega}^\top \mathbf{g} - \theta \mid \theta, \Sigma] = \boldsymbol{\omega}^\top \Sigma \boldsymbol{\omega}$ (see equation (1.5.3), pp. 14 of Mardia et al. (1979)).

To obtain the posterior expectation of θ , define $\boldsymbol{\omega}^*$ as the vector of weights for which the linear combination generates the unbiased posterior expectation. When Σ is positive definite (PD) and invertible, we can compute $\boldsymbol{\omega}^*$ by solving the following constrained minimization problem:

$$\boldsymbol{\omega}^* \in \underset{\boldsymbol{\omega} \in \mathbb{R}^p}{\text{argmin}} \boldsymbol{\omega}^\top \Sigma \boldsymbol{\omega} \quad \text{s.t.} \quad \mathbf{e}^\top \boldsymbol{\omega} = 1. \quad (\text{EC.2})$$

We can construct the following Lagrangian function: $\mathcal{L}(\boldsymbol{\omega}, \tau) = \boldsymbol{\omega}^\top \Sigma \boldsymbol{\omega} + \tau(1 - \mathbf{e}^\top \boldsymbol{\omega})$ and impose the first-order condition by setting the gradient with respect to $\boldsymbol{\omega}$ to $\mathbf{0}$. This gives $\nabla_{\boldsymbol{\omega}} \mathcal{L}(\boldsymbol{\omega}, \tau) = 2\Sigma \boldsymbol{\omega} - \tau \mathbf{e} = \mathbf{0}$ (since the covariance matrix Σ is by definition symmetric), and therefore $2\Sigma \boldsymbol{\omega} = \tau \mathbf{e}$,

which yields $\boldsymbol{\omega} = \frac{\tau}{2} \Sigma^{-1} \mathbf{e}$. To obtain the value of the Lagrangian multiplier, we can impose the equality constraint: $1 = \mathbf{e}^\top \boldsymbol{\omega} = \frac{\tau}{2} \mathbf{e}^\top \Sigma^{-1} \mathbf{e}$, which yields $\tau = \frac{2}{\mathbf{e}^\top \Sigma^{-1} \mathbf{e}}$. Finally this gives the optimal vector of weights $\boldsymbol{\omega}^* = \frac{\Sigma^{-1} \mathbf{e}}{\mathbf{e}^\top \Sigma^{-1} \mathbf{e}}$, and subsequently $\boldsymbol{\omega}^{*\top} \mathbf{g}$ is the minimum-variance linear unbiased estimator (MVLUE) of θ when Σ is known. As pointed out previously, if the distribution of forecast errors belongs to the family of elliptically contoured distributions, then the MVLUE coincides with the minimum-variance unbiased estimator (MVUE) among all classes of estimators. (An astute reader may observe that the weights in equation (EC.1) exactly match the weights presented in equation (3.7) of Winkler (1981). Of note, the present derivation generalizes the result in Winkler (1981) to accommodate any elliptically contoured error distribution but does not describe the posterior distribution of θ beyond its first moment.) This completes the proof. $\square$

EC.1.2 Estimating the Borg Effect and Algorithm Aversion When the AI Advice is Known

For each historical forecasting task $i \in \{1, \dots, N\}$, let f_{ji}^0 , f_{ji}^1 , θ_i , f_i^a , and J_i be the initial forecast by expert j , the final forecast by expert j , the correct answer for task i , the AI advice for task i , and the number of forecasters in the crowd for task i , respectively. The extent of the borg effect $\hat{\beta}$ and the degree of algorithm aversion $\hat{\alpha}$ in the information environment can be estimated as follows:

$$\hat{\beta} \in \operatorname{argmin}_{\beta \in [0,1]} \sum_{i=1}^N (\beta \hat{w}^* f_i^a + (1 - \beta \hat{w}^*) \bar{f}_i^0 - \theta_i)^2; \quad (\text{EC.3})$$

$$\hat{\alpha} \in \operatorname{argmin}_{\alpha \geq 0} \sum_{i=1}^N \sum_{j=1}^{J_i} \left(\frac{\hat{w}_i}{\alpha} f_i^a + \left(1 - \frac{\hat{w}_i}{\alpha} \right) f_{ji}^0 + \hat{\varepsilon}_{ji}^1 - \theta_i \right)^2; \quad (\text{EC.4})$$

where $\hat{w}^* \in \operatorname{argmin}_{w \in [0,1]} \sum_{i=1}^N \sum_{j=1}^{J_i} (w f_i^a + (1 - w) f_{ji}^0 - \theta_i)^2$, $\hat{w}_i \in \operatorname{argmin}_{w_i \in [0,1]} \sum_{j=1}^{J_i} (w_i f_i^a + (1 - w_i) f_{ji}^0 - f_{ji}^1)^2$, and $\hat{\varepsilon}_{ji}^1 = f_{ji}^1 - (\hat{w}_i f_i^a + (1 - \hat{w}_i) f_{ji}^0)$, for all $i \in \{1, \dots, N\}$, $j \in \{1, \dots, J_i\}$.

To estimate the borg effect and algorithm aversion in the crowd using training data, we follow the principle of empirical risk minimization. To estimate the borg effect as $\hat{\beta}$, we first empirically estimate the optimal weight on advice $\hat{w}^*$ that individual forecasters across all training tasks should have used in order to maximize the accuracy of their individual final forecasts. When $\hat{w}^* \in [0, 1]$, it can be calculated as the slope of the no-intercept OLS regression $(f_{ji}^0 - \theta_i) = w^* (f_{ji}^0 - f_i^a) + \chi_{ji}$. Finally, we compute $\hat{\beta}$ as the amount by which the empirically-optimal individual weight on advice $\hat{w}^*$ should be downweighted to minimize the squared error of aggregate final forecasts.

To estimate the algorithm aversion as $\hat{\alpha}$, we first calculate the observed average weight on advice $\hat{w}_i$ for the i -th task, which can most closely produce the set of observed final forecasts f_{ji}^1 over J_i pairs of responses. Of note, when $\hat{w}_i \in [0, 1]$, it can be equivalently calculated as the coefficient of the no-intercept OLS regression $(f_{ji}^1 - f_{ji}^0) = w_i (f_i^a - f_{ji}^0) + \varepsilon_{ji}^1$. Next, we create task-specific

counterfactual crowds of rational experts who on average use an empirically-optimal, error variance-minimizing weight on advice. If the crowd has an average degree of algorithm aversion of α , then this counterfactual average weight on advice is given by $\frac{\hat{w}_i}{\alpha}$, since $w = \alpha w^*$. In addition to using this ideal weight on advice, we also assume that judge j 's unbiased final forecast would continue to have the observed residual of $\hat{\varepsilon}_{ji}^1$. These yield judge j 's individual counterfactual final forecast to be $\frac{\hat{w}_i}{\alpha} f_i^a + \left(1 - \frac{\hat{w}_i}{\alpha}\right) f_{ji}^0 + \hat{\varepsilon}_{ji}^1$. Finally, we estimate $\hat{\alpha}$ as the degree of algorithm aversion which produces a set of counterfactual final forecasts that most closely align with individually-rational final forecasts, by minimizing the squared error of individual AI-assisted counterfactual forecasts.

EC.2 Proofs of Propositions in the Main Paper

EC.2.1 Proof of Proposition 1

To derive the individual posterior expectation for the j -th judge using the vector of forecasts $\mathbf{g}_\eta = \{f^a, f_j^0\}$, we need to calculate $\boldsymbol{\omega}_\eta^* = \{\eta_a, \eta_0\}$ from equation (EC.1) in Lemma EC.1. Define the following 2×2 forecast error covariance matrix:

$$\Sigma_\eta = \begin{bmatrix} \text{Var}[f^a - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] & \text{Cov}[f^a - \theta, f_j^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] \\ \text{Cov}[f^a - \theta, f_j^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] & \text{Var}[f_j^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] \end{bmatrix}.$$

Now observe that

$$\text{Var}[f^a - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] = \text{Var}[s + x | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] = \sigma_s^2 + \sigma_x^2, \quad (\text{EC.5})$$

$$\text{Var}[f_j^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] = \text{Var}[s + y + \varepsilon_j^0 | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] = \sigma_s^2 + \sigma_y^2 + \sigma_0^2, \quad (\text{EC.6})$$

$$\text{Cov}[f^a - \theta, f_j^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] = \text{Cov}[s + x, s + y + \varepsilon_j^0 | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] = \sigma_s^2, \quad (\text{EC.7})$$

for all $j \in \{1, \dots, J\}$. These yield

$$\Sigma_\eta = \begin{bmatrix} \sigma_s^2 + \sigma_x^2 & \sigma_s^2 \\ \sigma_s^2 & \sigma_s^2 + \sigma_y^2 + \sigma_0^2 \end{bmatrix}.$$

In this proof, denote $\mathbf{e}$ as the unit vector of length 2. First we write $\Sigma_\eta = \tilde{\Sigma}_\eta + \sigma_s^2 \mathbf{e} \mathbf{e}^\top$, where $\tilde{\Sigma}_\eta = \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 + \sigma_0^2 \end{bmatrix}$. Observe that $\tilde{\Sigma}_\eta$ is PD when $\sigma_x^2, \sigma_y^2, \sigma_0^2 > 0$, and $\sigma_s^2 \mathbf{e} \mathbf{e}^\top$ is positive semi-definite (PSD) since $\det(\mathbf{e} \mathbf{e}^\top) = 0$. The sum of a PD matrix and PSD matrix is PD, and therefore Σ_η is PD and invertible.

Next, we use the Sherman–Morrison–Woodbury formula (Woodbury 1950, Sherman and Morrison 1950), to show that

$$\begin{aligned} \Sigma_\eta^{-1} \mathbf{e} &= \left(\tilde{\Sigma}_\eta + \sigma_s^2 \mathbf{e} \mathbf{e}^\top \right)^{-1} \mathbf{e} = \left(\tilde{\Sigma}_\eta^{-1} - \frac{\sigma_s^2 \tilde{\Sigma}_\eta^{-1} \mathbf{e} \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1}}{1 + \sigma_s^2 \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}} \right) \mathbf{e} \\ &= \tilde{\Sigma}_\eta^{-1} \mathbf{e} - \frac{\sigma_s^2 \tilde{\Sigma}_\eta^{-1} \mathbf{e} \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}}{1 + \sigma_s^2 \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}} = \frac{\tilde{\Sigma}_\eta^{-1} \mathbf{e} + \sigma_s^2 \tilde{\Sigma}_\eta^{-1} \mathbf{e} \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e} - \sigma_s^2 \tilde{\Sigma}_\eta^{-1} \mathbf{e} \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}}{1 + \sigma_s^2 \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}} = \frac{\tilde{\Sigma}_\eta^{-1} \mathbf{e}}{1 + \sigma_s^2 \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}}. \\ \Rightarrow \mathbf{e}^\top \Sigma_\eta^{-1} \mathbf{e} &= \frac{\mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}}{1 + \sigma_s^2 \mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}} \Rightarrow \frac{\Sigma_\eta^{-1} \mathbf{e}}{\mathbf{e}^\top \Sigma_\eta^{-1} \mathbf{e}} = \frac{\tilde{\Sigma}_\eta^{-1} \mathbf{e}}{\mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}}. \end{aligned} \quad (\text{EC.8})$$

It is straightforward to show that $\tilde{\Sigma}_\eta^{-1} = \frac{1}{\sigma_x^2(\sigma_y^2 + \sigma_0^2)} \begin{bmatrix} \sigma_y^2 + \sigma_0^2 & 0 \\ 0 & \sigma_x^2 \end{bmatrix}$, and from equation (EC.1) we get $\omega_\eta^* = \frac{\Sigma_\eta^{-1} \mathbf{e}}{\mathbf{e}^\top \Sigma_\eta^{-1} \mathbf{e}} = \frac{\tilde{\Sigma}_\eta^{-1} \mathbf{e}}{\mathbf{e}^\top \tilde{\Sigma}_\eta^{-1} \mathbf{e}} = \frac{1}{\frac{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}{\sigma_x^2(\sigma_y^2 + \sigma_0^2)}} \left\{ \frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2(\sigma_y^2 + \sigma_0^2)}, \frac{\sigma_x^2}{\sigma_x^2(\sigma_y^2 + \sigma_0^2)} \right\} = \left\{ \frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}, \frac{\sigma_x^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right\} = \{w^*, 1 - w^*\}$, where $w^* = \frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}$. This yields $f_j^{1*} = \omega_\eta^{*\top} \mathbf{g}_\eta = w^* f^a + (1 - w^*) f_j^0$, which completes the proof. $\square$

EC.2.2 Proof of Proposition 2

To derive the GPE using the vector of forecasts $\mathbf{g}_\zeta = \{f^a, f_1^0, \dots, f_J^0\}_{J+1}$, we need to calculate $\omega_\zeta^* = \{\zeta_a, \zeta_1, \dots, \zeta_J\}_{J+1}$ from equation (EC.1) in Lemma EC.1.

Define the following $(J + 1) \times (J + 1)$ forecast error covariance matrix:

$$\Sigma_\zeta = \begin{bmatrix} \text{Var}[f^a - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] & \text{Cov}[f^a - \theta, f_1^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] & \dots & \text{Cov}[f^a - \theta, f_J^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] \\ \text{Cov}[f_1^0 - \theta, f^a - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] & \text{Var}[f_1^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] & \dots & \text{Cov}[f_1^0 - \theta, f_J^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}[f_J^0 - \theta, f^a - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] & \text{Cov}[f_J^0 - \theta, f_1^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] & \dots & \text{Var}[f_J^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] \end{bmatrix}.$$

In addition to equations (EC.5), (EC.6), (EC.7), observe that

$$\text{Cov}[f_k^0 - \theta, f_j^0 - \theta | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] = \text{Cov}[s + y + \varepsilon_k^0, s + y + \varepsilon_j^0 | \theta, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2] = \sigma_s^2 + \sigma_y^2 \quad (\text{EC.9})$$

for all $k \neq j$; $j, k \in \{1, \dots, J\}$. These yield

$$\Sigma_\zeta = \begin{bmatrix} \sigma_s^2 + \sigma_x^2 & \sigma_s^2 & \dots & \sigma_s^2 \\ \sigma_s^2 & \sigma_s^2 + \sigma_y^2 + \sigma_0^2 & \dots & \sigma_s^2 + \sigma_y^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_s^2 & \sigma_s^2 + \sigma_y^2 & \dots & \sigma_s^2 + \sigma_y^2 + \sigma_0^2 \end{bmatrix}.$$

In this proof, denote $\mathbf{e}$ as the unit vector of length J and $\mathbf{I}$ as the $J \times J$ identity matrix. First, we write $\Sigma_\zeta = \tilde{\Sigma}_\zeta + \sigma_s^2 \mathbf{e} \mathbf{e}^\top$, $\tilde{\Sigma}_\zeta = \begin{bmatrix} \sigma_x^2 & \mathbf{0} \\ \mathbf{0} & \tilde{\Sigma}_\zeta \end{bmatrix}$ and $\tilde{\Sigma}_\zeta = \begin{bmatrix} \sigma_y^2 + \sigma_0^2 & \dots & \sigma_y^2 \\ \vdots & \ddots & \vdots \\ \sigma_y^2 & \dots & \sigma_y^2 + \sigma_0^2 \end{bmatrix} = \sigma_0^2 \mathbf{I} + \sigma_y^2 \mathbf{e} \mathbf{e}^\top$. Observe that $\tilde{\Sigma}_\zeta$ is PD (using the same argument as $\tilde{\Sigma}_\eta$). With $\sigma_x^2 > 0$, $\tilde{\Sigma}_\zeta$ is also PD since a diagonal block matrix is PD when each diagonal block is PD. If $\tilde{\Sigma}_\zeta$ is PD, Σ_ζ is also PD and invertible.

From the structure of Σ_ζ , we first use the Sherman-Morrison-Woodbury formula to show that

$$\frac{\Sigma_\zeta^{-1} \mathbf{e}}{\mathbf{e}^\top \Sigma_\zeta^{-1} \mathbf{e}} = \frac{\tilde{\Sigma}_\zeta^{-1} \mathbf{e}}{\mathbf{e}^\top \tilde{\Sigma}_\zeta^{-1} \mathbf{e}} \quad (\text{identical to Proposition 1}). \text{ We express } \tilde{\Sigma}_\zeta^{-1} = \begin{bmatrix} \frac{1}{\sigma_x^2} & \mathbf{0} \\ \mathbf{0} & \tilde{\Sigma}_\zeta^{-1} \end{bmatrix}, \text{ using the result}$$

on inversion of block matrices given in §2.6.1 of Press (2012). Observe that the off-diagonal block matrices in $\tilde{\Sigma}_\zeta$ are null, which also holds in its inverse. Using equation (2.5.9) from Press (2012) to calculate the inverse of an intra-class correlation matrix such as $\tilde{\Sigma}_\zeta$, we show that $\tilde{\Sigma}_\zeta^{-1} =$

$$(\sigma_0^2 \mathbf{I} + \sigma_y^2 \mathbf{e} \mathbf{e}^\top)^{-1} = \frac{\mathbf{I}}{\sigma_0^2} - \frac{\sigma_y^2 \mathbf{e} \mathbf{e}^\top}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)}. \text{ This yields } \tilde{\Sigma}_\zeta^{-1} = \begin{bmatrix} \frac{1}{\sigma_x^2} & \mathbf{0} \\ \mathbf{0} & \frac{\mathbf{I}}{\sigma_0^2} - \frac{\sigma_y^2 \mathbf{e} \mathbf{e}^\top}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)} \end{bmatrix}, \text{ and since}$$

$$\mathbf{e}^\top \tilde{\Sigma}_\zeta^{-1} \mathbf{e} \text{ gives the scalar sum of all elements of } \tilde{\Sigma}_\zeta^{-1}, \mathbf{e}^\top \tilde{\Sigma}_\zeta^{-1} \mathbf{e} = \frac{1}{\sigma_x^2} + \frac{J}{\sigma_0^2} - \frac{J^2 \sigma_y^2}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)} = \frac{1}{\sigma_x^2} + \frac{J^2 \sigma_y^2 + J \sigma_0^2 - J^2 \sigma_y^2}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)} = \frac{1}{\sigma_x^2} + \frac{J \sigma_0^2}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)} = \frac{1}{\sigma_x^2} + \frac{J}{J \sigma_y^2 + \sigma_0^2} = \frac{1}{\sigma_x^2} + \frac{1}{\sigma_y^2 + \frac{\sigma_0^2}{J}} = \frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}.$$

$$\begin{aligned} \text{From equation (EC.1), } \omega_\zeta^* &= \frac{\tilde{\Sigma}_\zeta^{-1} \mathbf{e}}{\mathbf{e}^\top \tilde{\Sigma}_\zeta^{-1} \mathbf{e}} = \frac{1}{\frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}} \left\{ \frac{1}{\sigma_x^2}, \frac{1}{\sigma_0^2} - \frac{J \sigma_y^2}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)}, \dots, \frac{1}{\sigma_0^2} - \frac{J \sigma_y^2}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)} \right\} \\ &= \frac{1}{\frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}} \left\{ \frac{1}{\sigma_x^2}, \frac{(J \sigma_y^2 + \sigma_0^2) - J \sigma_y^2}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)}, \dots, \frac{(J \sigma_y^2 + \sigma_0^2) - J \sigma_y^2}{\sigma_0^2 (J \sigma_y^2 + \sigma_0^2)} \right\} = \frac{1}{\frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}} \left\{ \frac{1}{\sigma_x^2}, \frac{1}{J \sigma_y^2 + \frac{\sigma_0^2}{J}}, \dots, \frac{1}{J \sigma_y^2 + \frac{\sigma_0^2}{J}} \right\} = \\ &\left\{ \frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}, \frac{1}{J} \frac{\sigma_x^2}{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}, \dots, \frac{1}{J} \frac{\sigma_x^2}{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}} \right\} = \left\{ \beta w^*, \frac{1 - \beta w^*}{J}, \dots, \frac{1 - \beta w^*}{J} \right\}, \quad \text{where} \\ \beta &= \frac{\frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2}}{\frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}}. \text{ Finally, we get } \theta^* = \omega_\zeta^{*\top} \mathbf{g}_\zeta = \beta w^* f^a + \frac{1 - \beta w^*}{J} f_1^0 + \dots + \frac{1 - \beta w^*}{J} f_J^0. \text{ We can rewrite} \end{aligned}$$

$$\text{this as } \theta^* = \beta w^* f^a + (1 - \beta w^*) \frac{1}{J} \sum_{j=1}^J f_j^0 = \beta w^* f^a + (1 - \beta w^*) \bar{f}^0.$$

In addition, observe that $0 < \sigma_y^2 + \frac{\sigma_0^2}{J} < \sigma_y^2 + \sigma_0^2 \Rightarrow 0 < \frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2} < 1$, and $\sigma_x^2 > 0$. We can show that

$$\frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} = \frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2} \cdot \frac{\frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \frac{\sigma_0^2}{J}}}{\frac{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}{\sigma_y^2 + \sigma_0^2}} = \frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2} \cdot \frac{1 + \frac{\sigma_x^2}{\sigma_y^2 + \frac{\sigma_0^2}{J}}}{1 + \frac{\sigma_x^2}{\sigma_y^2 + \sigma_0^2}} > \frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2} \cdot \frac{1 + \frac{\sigma_x^2}{\sigma_y^2 + \sigma_0^2}}{1 + \frac{\sigma_x^2}{\sigma_y^2 + \sigma_0^2}} = \frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2}. \text{ This yields } \frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} > \frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2} > 0, \text{ and therefore } 0 < \beta = \frac{\frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_y^2 + \sigma_0^2}}{\frac{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}} < 1, \text{ which completes the proof. } \square$$

EC.2.3 Proof of Proposition 3

First, we calculate the expected squared forecast errors of $\bar{f}^0$ and $\bar{f}^1$:

$$\mathcal{E}(\bar{f}^0) = \mathbb{E}[(\bar{f}^0 - \theta)^2 | \theta] = \mathbb{E} \left[\left(\theta + s + y + \frac{1}{J} \sum_{j=1}^J \varepsilon_j^0 - \theta \right)^2 \middle| \theta \right] = \mathbb{E} \left[\left(s + y + \frac{1}{J} \sum_{j=1}^J \varepsilon_j^0 \right)^2 \middle| \theta \right] = \sigma_s^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}; \quad (\text{EC.10})$$

$$\begin{aligned} \mathcal{E}(\bar{f}^1) &= \mathbb{E}[(\bar{f}^1 - \theta)^2 | \theta] = \mathbb{E} \left[\left(w(\theta + s + x) + (1 - w) \left(\theta + s + y + \frac{1}{J} \sum_{j=1}^J \varepsilon_j^0 \right) + \frac{1}{J} \sum_{j=1}^J \varepsilon_j^1 - \theta \right)^2 \middle| \theta \right] \\ &= \mathbb{E} \left[\left(s + wx + (1 - w) \left(y + \frac{1}{J} \sum_{j=1}^J \varepsilon_j^0 \right) + \frac{1}{J} \sum_{j=1}^J \varepsilon_j^1 \right)^2 \middle| \theta \right] = \sigma_s^2 + w^2 \sigma_x^2 + (1 - w)^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) + \frac{\sigma_1^2}{J}. \end{aligned} \quad (\text{EC.11})$$

Therefore we can show:

$$\begin{aligned}
\mathcal{E}(\bar{f}^0) - \mathcal{E}(\bar{f}^1) &= \sigma_s^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} - \sigma_s^2 - w^2 \sigma_x^2 - (1-w)^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) - \frac{\sigma_1^2}{J} \\
&= (1 - (1-w)^2) \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) - w^2 \sigma_x^2 - \frac{\sigma_1^2}{J} = (2w - w^2) \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) - w^2 \sigma_x^2 - \frac{\sigma_1^2}{J} \\
&= w \left((2-w) \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) - w \sigma_x^2 \right) - \frac{\sigma_1^2}{J} = w \left(2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) - w \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) \right) - \frac{\sigma_1^2}{J} \\
&= \alpha w^* \left(2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) - \alpha w^* \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) \right) - \frac{\sigma_1^2}{J} \quad (\text{since } w = \alpha w^*) \\
&= \alpha w^* \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) \left(2 \frac{\sigma_y^2 + \frac{\sigma_0^2}{J}}{\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J}} - \alpha w^* \right) - \frac{\sigma_1^2}{J} \\
&= \alpha w^* \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) (2\beta w^* - \alpha w^*) - \frac{\sigma_1^2}{J} = (2\beta - \alpha) \alpha w^{*2} \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) - \frac{\sigma_1^2}{J}. \\
&\Rightarrow \lim_{J \rightarrow \infty} \mathcal{E}(\bar{f}^0) - \mathcal{E}(\bar{f}^1) = (2\beta - \alpha) \alpha w^{*2} (\sigma_x^2 + \sigma_y^2),
\end{aligned}$$

and that completes the proof. $\square$

EC.2.4 Proof of Proposition 4

To derive the GPE estimate using the vector of forecasts $\mathbf{g}_\psi = \{f_1^0, \dots, f_J^0, f_1^1, \dots, f_J^1\}_{2J}$, we need to calculate $\boldsymbol{\omega}_\psi^* = \{\psi_1^0, \dots, \psi_J^0, \psi_1^1, \dots, \psi_J^1\}_{2J}$ from equation (EC.1) in Lemma EC.1. Define the following $2J \times 2J$ forecast error covariance matrix:

$$\Sigma_\psi = \begin{bmatrix} \left[\Sigma_\psi^{0,0} \right]_{J \times J} & \left[\Sigma_\psi^{0,1} \right]_{J \times J} \\ \left[\Sigma_\psi^{1,0} \right]_{J \times J} & \left[\Sigma_\psi^{1,1} \right]_{J \times J} \end{bmatrix},$$

where

$$\begin{aligned}
\Sigma_\psi^{0,0} &= \begin{bmatrix} \text{Var} [f_1^0 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] & \cdots & \text{Cov} [f_1^0 - \theta, f_J^0 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\ \vdots & \ddots & \vdots \\ \text{Cov} [f_J^0 - \theta, f_1^0 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] & \cdots & \text{Var} [f_J^0 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \end{bmatrix}_{J \times J}, \\
\Sigma_\psi^{1,1} &= \begin{bmatrix} \text{Var} [f_1^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] & \cdots & \text{Cov} [f_1^1 - \theta, f_J^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\ \vdots & \ddots & \vdots \\ \text{Cov} [f_J^1 - \theta, f_1^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] & \cdots & \text{Var} [f_J^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \end{bmatrix}_{J \times J}, \\
\Sigma_\psi^{0,1} &= \begin{bmatrix} \text{Cov} [f_1^0 - \theta, f_1^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] & \cdots & \text{Cov} [f_1^0 - \theta, f_J^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\ \vdots & \ddots & \vdots \\ \text{Cov} [f_J^0 - \theta, f_1^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] & \cdots & \text{Cov} [f_J^0 - \theta, f_J^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \end{bmatrix}_{J \times J},
\end{aligned}$$

and $\Sigma_\psi^{1,0} = (\Sigma_\psi^{0,1})^\top$. In addition to equations (EC.5), (EC.6), (EC.7), (EC.9), observe that

$$\begin{aligned}
\text{Var} [f_j^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] &= \text{Var} [w f^a + (1-w) f_j^0 + \varepsilon_j^1 - \theta \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= \text{Var} [w(f^a - \theta) + (1-w)(f_j^0 - \theta) + \varepsilon_j^1 \mid \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2]
\end{aligned}$$

$$\begin{aligned}
&= \text{Var} [w(s+x) + (1-w)(s+y+\varepsilon_j^0) + \varepsilon_j^1 | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= w^2(\sigma_s^2 + \sigma_x^2) + (1-w)^2(\sigma_s^2 + \sigma_y^2 + \sigma_0^2) + \sigma_1^2 + 2w(1-w)\sigma_s^2 \\
&= \sigma_s^2 + w^2\sigma_x^2 + (1-w)^2(\sigma_y^2 + \sigma_0^2) + \sigma_1^2,
\end{aligned} \tag{EC.12}$$

$$\begin{aligned}
\text{Cov} [f_j^1 - \theta, f_j^0 - \theta | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] &= \text{Cov} [wf^a + (1-w)f_j^0 + \varepsilon_j^1 - \theta, f_j^0 - \theta | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= \text{Cov} [w(f^a - \theta) + (1-w)(f_j^0 - \theta) + \varepsilon_j^1, f_j^0 - \theta | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= \text{Cov} [w(s+x) + (1-w)(s+y+\varepsilon_j^0) + \varepsilon_j^1, s+y+\varepsilon_j^0 | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= w\sigma_s^2 + (1-w)(\sigma_s^2 + \sigma_y^2 + \sigma_0^2) = \sigma_s^2 + (1-w)(\sigma_y^2 + \sigma_0^2),
\end{aligned} \tag{EC.13}$$

$$\begin{aligned}
\text{Cov} [f_j^0 - \theta, f_k^1 - \theta | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] &= \text{Cov} [f_j^0 - \theta, wf^a + (1-w)f_k^0 + \varepsilon_k^1 - \theta | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= \text{Cov} [f_j^0 - \theta, w(f^a - \theta) + (1-w)(f_k^0 - \theta) + \varepsilon_k^1 | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= \text{Cov} [s+y+\varepsilon_j^0, w(s+x) + (1-w)(s+y+\varepsilon_k^0) + \varepsilon_k^1 | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= w\sigma_s^2 + (1-w)(\sigma_s^2 + \sigma_y^2) = \sigma_s^2 + (1-w)\sigma_y^2,
\end{aligned} \tag{EC.14}$$

$$\begin{aligned}
\text{Cov} [f_j^1 - \theta, f_k^1 - \theta | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] &= \text{Cov} [wf^a + (1-w)f_j^0 + \varepsilon_j^1 - \theta, wf^a + (1-w)f_k^0 + \varepsilon_k^1 - \theta | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= \text{Cov} [w(f^a - \theta) + (1-w)(f_j^0 - \theta) + \varepsilon_j^1, w(f^a - \theta) + (1-w)(f_k^0 - \theta) + \varepsilon_k^1 | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= \text{Cov} [w(s+x) + (1-w)(s+y+\varepsilon_j^0) + \varepsilon_j^1, w(s+x) + (1-w)(s+y+\varepsilon_k^0) + \varepsilon_k^1 | \theta, w, \sigma_s^2, \sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2] \\
&= w^2(\sigma_s^2 + \sigma_x^2) + 2w(1-w)\sigma_s^2 + (1-w)^2(\sigma_s^2 + \sigma_y^2) = \sigma_s^2 + w^2\sigma_x^2 + (1-w)^2\sigma_y^2,
\end{aligned} \tag{EC.15}$$

for all $k \neq j$; $j, k \in \{1, \dots, J\}$. These yield

$$\begin{aligned}
\Sigma_\psi^{0,0} &= \begin{bmatrix} \sigma_s^2 + \sigma_y^2 + \sigma_0^2 & \cdots & \sigma_s^2 + \sigma_y^2 \\ \vdots & \ddots & \vdots \\ \sigma_s^2 + \sigma_y^2 & \cdots & \sigma_s^2 + \sigma_y^2 + \sigma_0^2 \end{bmatrix}, \\
\Sigma_\psi^{1,1} &= \begin{bmatrix} \sigma_s^2 + w^2\sigma_x^2 + (1-w)^2(\sigma_y^2 + \sigma_0^2) + \sigma_1^2 & \cdots & \sigma_s^2 + w^2\sigma_x^2 + (1-w)^2\sigma_y^2 \\ \vdots & \ddots & \vdots \\ \sigma_s^2 + w^2\sigma_x^2 + (1-w)^2\sigma_y^2 & \cdots & \sigma_s^2 + w^2\sigma_x^2 + (1-w)^2(\sigma_y^2 + \sigma_0^2) + \sigma_1^2 \end{bmatrix}, \\
\Sigma_\psi^{0,1} = \Sigma_\psi^{1,0} &= \begin{bmatrix} \sigma_s^2 + (1-w)(\sigma_y^2 + \sigma_0^2) & \cdots & \sigma_s^2 + (1-w)\sigma_y^2 \\ \vdots & \ddots & \vdots \\ \sigma_s^2 + (1-w)\sigma_y^2 & \cdots & \sigma_s^2 + (1-w)(\sigma_y^2 + \sigma_0^2) \end{bmatrix}.
\end{aligned}$$

In this proof, denote $\mathbf{e}_J$ as the unit vector of length J and $\mathbf{I}_J$ as the $J \times J$ identity matrix. First, we write $\Sigma_\psi^{0,0} = \tilde{\Sigma}_\psi^{0,0} + (\sigma_s^2 + \sigma_y^2)\mathbf{e}_J\mathbf{e}_J^\top$, $\Sigma_\psi^{1,1} = \tilde{\Sigma}_\psi^{1,1} + (\sigma_s^2 + w^2\sigma_x^2 + (1-w)^2\sigma_y^2)\mathbf{e}_J\mathbf{e}_J^\top$, $\Sigma_\psi^{0,1} =$

$\tilde{\Sigma}_{\psi}^{0,1} + (\sigma_s^2 + (1-w)\sigma_y^2) \mathbf{e}_J \mathbf{e}_J^\top$, and $\tilde{\Sigma}_{\psi}^{1,0} = \tilde{\Sigma}_{\psi}^{1,0} + (\sigma_s^2 + (1-w)\sigma_y^2) \mathbf{e}_J \mathbf{e}_J^\top$, where $\tilde{\Sigma}_{\psi}^{0,0} = \sigma_0^2 \mathbf{I}_J$, $\tilde{\Sigma}_{\psi}^{1,1} = ((1-w)^2\sigma_0^2 + \sigma_1^2) \mathbf{I}_J$, and $\tilde{\Sigma}_{\psi}^{0,1} = \tilde{\Sigma}_{\psi}^{1,0} = (1-w)\sigma_0^2 \mathbf{I}_J$. We argue that if $\sigma_x^2, \sigma_y^2, \sigma_0^2, \sigma_1^2 > 0$, then $\tilde{\Sigma}_{\psi}^{0,0}$, $\tilde{\Sigma}_{\psi}^{1,1}$, $\tilde{\Sigma}_{\psi}^{0,1}$, $\tilde{\Sigma}_{\psi}^{1,0}$ are all PD. It follows that a block matrix where every diagonal block is PD is itself PD, and therefore Σ_{ψ} is PD and invertible.

Although it is easy to calculate the inverse of each individual $J \times J$ intra-class correlation matrix in Σ_{ψ} , combining the four inverted matrices to determine the full Σ_{ψ}^{-1} matrix is algebraically challenging since the off-diagonal blocks are not null. To allow for a simpler analysis, in equation (EC.16) we decompose Σ_{ψ} into a low-dimensional Kronecker sum-product form:

$$\Sigma_{\psi} = S_0 \otimes \mathbf{I}_J + S_1 \otimes (\mathbf{e}_J \mathbf{e}_J^\top), \quad (\text{EC.16})$$

where $S_0 = \begin{bmatrix} \sigma_0^2 & (1-w)\sigma_0^2 \\ (1-w)\sigma_0^2 & (1-w)^2\sigma_0^2 + \sigma_1^2 \end{bmatrix}_{2 \times 2}$, $S_1 = \begin{bmatrix} \sigma_s^2 + \sigma_y^2 & \sigma_s^2 + (1-w)\sigma_y^2 \\ \sigma_s^2 + (1-w)\sigma_y^2 & \sigma_s^2 + w^2\sigma_x^2 + (1-w)^2\sigma_y^2 \end{bmatrix}_{2 \times 2}$, and $\otimes$ is the symbol for the Kronecker product (Press (2012), equation (2.10.1)). To elaborate, the

Kronecker product of two matrices $\mathbf{A}_{m_1 \times n_1}$ and $\mathbf{B}_{m_2 \times n_2}$ is given by $\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & \cdots & a_{1n_1}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m_1 1}\mathbf{B} & \cdots & a_{m_1 n_1}\mathbf{B} \end{bmatrix}$

and is of dimension of $m_1 m_2 \times n_1 n_2$. Importantly, equation (EC.16) allows us to differentiate between within-group variation ($S_0 \otimes \mathbf{I}_J$) and between-group variation ($S_1 \otimes (\mathbf{e}_J \mathbf{e}_J^\top)$) that is contained within Σ_{ψ} . Note that S_0 and S_1 are both PD and invertible because $\sigma_1^2 > 0$ and $\sigma_x^2 > 0$, respectively. Denote $\mathbf{I}_2$ as the 2×2 identity matrix. We demonstrate the following:

$$\begin{aligned} \Sigma_{\psi}^{-1} &= (S_0 \otimes \mathbf{I}_J + S_1 \otimes (\mathbf{e}_J \mathbf{e}_J^\top))^{-1} = (S_0 \otimes \mathbf{I}_J + (S_1 \mathbf{I}_2) \otimes (\mathbf{e}_J \mathbf{e}_J^\top))^{-1} \\ &= (S_0 \otimes \mathbf{I}_J + (S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 \otimes \mathbf{e}_J^\top))^{-1} \quad (\text{Press (2012), equation (2.10.3)}) \\ &= (S_0 \otimes \mathbf{I}_J + (S_1 \otimes \mathbf{e}_J) \mathbf{I}_2 (\mathbf{I}_2 \otimes \mathbf{e}_J^\top))^{-1} \quad (\text{Sherman-Morrison-Woodbury form}) \\ &= (S_0 \otimes \mathbf{I}_J)^{-1} \\ &\quad - (S_0 \otimes \mathbf{I}_J)^{-1} (S_1 \otimes \mathbf{e}_J) \mathbf{I}_2 \left(\mathbf{I}_2 + \mathbf{I}_2 (\mathbf{I}_2 \otimes \mathbf{e}_J^\top) (S_0 \otimes \mathbf{I}_J)^{-1} (S_1 \otimes \mathbf{e}_J) \mathbf{I}_2 \right)^{-1} \mathbf{I}_2 (\mathbf{I}_2 \otimes \mathbf{e}_J^\top) (S_0 \otimes \mathbf{I}_J)^{-1} \\ &\quad (\text{Press (2012), equation (2.5.5)}) \\ &= (S_0 \otimes \mathbf{I}_J)^{-1} - (S_0 \otimes \mathbf{I}_J)^{-1} (S_1 \otimes \mathbf{e}_J) \left(\mathbf{I}_2 + (\mathbf{I}_2 \otimes \mathbf{e}_J^\top) (S_0 \otimes \mathbf{I}_J)^{-1} (S_1 \otimes \mathbf{e}_J) \right)^{-1} (\mathbf{I}_2 \otimes \mathbf{e}_J^\top) (S_0 \otimes \mathbf{I}_J)^{-1} \\ &= S_0^{-1} \otimes \mathbf{I}_J - (S_0^{-1} \otimes \mathbf{I}_J) (S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + (\mathbf{I}_2 \otimes \mathbf{e}_J^\top) (S_0^{-1} \otimes \mathbf{I}_J) (S_1 \otimes \mathbf{e}_J))^{-1} (\mathbf{I}_2 \otimes \mathbf{e}_J^\top) (S_0^{-1} \otimes \mathbf{I}_J) \\ &\quad (\text{Press (2012), equation (2.10.3), and } \mathbf{I}_J = \mathbf{I}_J^{-1}) \\ &= S_0^{-1} \otimes \mathbf{I}_J - (S_0^{-1} S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + (\mathbf{I}_2 \otimes \mathbf{e}_J^\top) (S_0^{-1} S_1) \otimes \mathbf{e}_J)^{-1} (S_0^{-1} \otimes \mathbf{e}_J^\top) \\ &= S_0^{-1} \otimes \mathbf{I}_J - (S_0^{-1} S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + (S_0^{-1} S_1) \otimes (\mathbf{e}_J^\top \mathbf{e}_J))^{-1} (S_0^{-1} \otimes \mathbf{e}_J^\top) \\ &= S_0^{-1} \otimes \mathbf{I}_J - (S_0^{-1} S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + J S_0^{-1} S_1)^{-1} (S_0^{-1} \otimes \mathbf{e}_J^\top). \end{aligned}$$

Denote $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_{2J}$ as unit vectors of length 1, 2, and $2J$ respectively. Note that $\mathbf{e}_{2J} = \mathbf{e}_2 \otimes \mathbf{e}_J$. Accordingly, we can show

$$\Sigma_{\psi}^{-1} \mathbf{e}_{2J} = \left(S_0^{-1} \otimes \mathbf{I}_J - (S_0^{-1} S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + J S_0^{-1} S_1)^{-1} (S_0^{-1} \otimes \mathbf{e}_J^\top) \right) \mathbf{e}_{2J}$$

$$\begin{aligned}
&= \left(S_0^{-1} \otimes \mathbf{I}_J - (S_0^{-1} S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + JS_0^{-1} S_1)^{-1} (S_0^{-1} \otimes \mathbf{e}_J^\top) \right) (\mathbf{e}_2 \otimes \mathbf{e}_J) \\
&= (S_0^{-1} \otimes \mathbf{I}_J) (\mathbf{e}_2 \otimes \mathbf{e}_J) - (S_0^{-1} S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + JS_0^{-1} S_1)^{-1} (S_0^{-1} \otimes \mathbf{e}_J^\top) (\mathbf{e}_2 \otimes \mathbf{e}_J) \\
&= (S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J - (S_0^{-1} S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + JS_0^{-1} S_1)^{-1} (S_0^{-1} \mathbf{e}_2 \otimes \mathbf{e}_J^\top \mathbf{e}_J) \\
&= (S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J - (S_0^{-1} S_1 \otimes \mathbf{e}_J) (\mathbf{I}_2 + JS_0^{-1} S_1)^{-1} (JS_0^{-1} \mathbf{e}_2).
\end{aligned}$$

Let $\mathbf{M} := \mathbf{I}_2 + JS_0^{-1} S_1 \Rightarrow S_0^{-1} S_1 = \frac{\mathbf{M} - \mathbf{I}_2}{J}$. This gives

$$\begin{aligned}
\Sigma_\psi^{-1} \mathbf{e}_{2J} &= (S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J - \left(\frac{\mathbf{M} - \mathbf{I}_2}{J} \right) \otimes \mathbf{e}_J \mathbf{M}^{-1} (JS_0^{-1} \mathbf{e}_2) = (S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J - ((\mathbf{M} - \mathbf{I}_2) \otimes \mathbf{e}_J) \mathbf{M}^{-1} (S_0^{-1} \mathbf{e}_2) \\
&= (S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J - (\mathbf{M} \otimes \mathbf{e}_J) \mathbf{M}^{-1} (S_0^{-1} \mathbf{e}_2) + (\mathbf{I}_2 \otimes \mathbf{e}_J) \mathbf{M}^{-1} (S_0^{-1} \mathbf{e}_2) \\
&= (S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J - (\mathbf{M} \otimes \mathbf{e}_J) (\mathbf{M}^{-1} S_0^{-1} \mathbf{e}_2 \otimes \mathbf{e}_1) + (\mathbf{I}_2 \otimes \mathbf{e}_J) (\mathbf{M}^{-1} S_0^{-1} \mathbf{e}_2 \otimes \mathbf{e}_1) \\
&= (S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J - (S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J + (\mathbf{M}^{-1} S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J = (\mathbf{M}^{-1} S_0^{-1} \mathbf{e}_2) \otimes \mathbf{e}_J = \mathbf{m} \otimes \mathbf{e}_J,
\end{aligned}$$

where $\mathbf{m} := \mathbf{M}^{-1} S_0^{-1} \mathbf{e}_2$. Finally we can show that $\mathbf{e}_{2J}^\top \Sigma_\psi^{-1} \mathbf{e}_{2J} = (\mathbf{e}_2 \otimes \mathbf{e}_J)^\top (\mathbf{m} \otimes \mathbf{e}_J) = (\mathbf{e}_2^\top \otimes \mathbf{e}_J^\top) (\mathbf{m} \otimes \mathbf{e}_J) = (\mathbf{e}_2^\top \mathbf{m}) \otimes (\mathbf{e}_J^\top \mathbf{e}_J) = J \mathbf{e}_2^\top \mathbf{m}$, using Press (2012), equation (2.10.6). Therefore from equation (EC.1),

$$\omega_\psi^* = \frac{\mathbf{m} \otimes \mathbf{e}_J}{J \mathbf{e}_2^\top \mathbf{m}}. \quad (\text{EC.17})$$

Now define $\mathbf{m} := \{m_1, m_2\}$, which implies $J \mathbf{e}_2^\top \mathbf{m}$ in the denominator is a scalar normalizing constant. Also note that in the numerator, $\mathbf{m} \otimes \mathbf{e}_J = \{m_1, m_2\} \otimes \mathbf{e}_J = \underbrace{\{m_1, \dots, m_1\}}_{J \text{ times}} \otimes \underbrace{\{m_2, \dots, m_2\}}_{J \text{ times}}$.

This shows that every initial forecast f_j^0 will receive identical weight $\frac{m_1}{J(m_1 + m_2)}$ and every final forecast f_j^1 will receive identical weight $\frac{m_2}{J(m_1 + m_2)}$. It is straightforward to evaluate $\mathbf{m}$ as a

$$\frac{m_1}{J(m_1 + m_2)} = \frac{1}{J} \left(1 - \frac{w \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}{w^2 \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) + \frac{\sigma_1^2}{J}} \right) \quad \text{and} \quad \frac{m_2}{J(m_1 + m_2)} = \frac{1}{J} \frac{w \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}{w^2 \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) + \frac{\sigma_1^2}{J}}.$$

Finally by substituting $w = \alpha w^* = \alpha \frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}$, we get the optimal weights $\omega_\psi^* =$

$$\left\{ \frac{\gamma^*}{J}, \dots, \frac{\gamma^*}{J}, \frac{1 - \gamma^*}{J}, \dots, \frac{1 - \gamma^*}{J} \right\}, \quad \text{where } \gamma^* = 1 - \frac{\alpha \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right) \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}{\alpha^2 \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right)^2 \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) + \frac{\sigma_1^2}{J}}.$$

$$\begin{aligned}
\hat{\theta}^* &= \omega_\psi^{*\top} \mathbf{g}_\psi = \frac{\gamma^*}{J} f_1^0 + \dots + \frac{\gamma^*}{J} f_J^0 + \frac{1 - \gamma^*}{J} f_1^1 + \dots + \frac{1 - \gamma^*}{J} f_J^1. \quad \text{We can rewrite this as } \hat{\theta}^* = \\
&\gamma^* \frac{1}{J} \sum_{j=1}^J f_j^0 + (1 - \gamma^*) \frac{1}{J} \sum_{j=1}^J f_j^1 = \gamma^* \bar{f}^0 + (1 - \gamma^*) \bar{f}^1 = \bar{f}^1 + \gamma^* (\bar{f}^0 - \bar{f}^1).
\end{aligned}$$

Observe that $\sup \gamma^* = 1$ (when $\alpha \rightarrow 0^+$ and σ_1^2 does not vanish), and $\inf \gamma^* = -\infty$ (when $\alpha, \sigma_x^2, \sigma_1^2 \rightarrow 0^+$). Therefore we can provide unconditional limits of the optimal pivoting factor as $\gamma^* \in (-\infty, 1)$, which completes the proof. $\square$

EC.2.5 Proof of Proposition 5

First, we demonstrate the representation of the asymptotically optimal pivoting factor as a function of β and α . From equation (10), $\gamma^* = 1 - \frac{\alpha \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right) \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}{\alpha^2 \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right)^2 \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right) + \frac{\sigma_1^2}{J}}$. Since $\lim_{J \rightarrow \infty} \frac{\sigma_1^2}{J} = 0$, we can show $\lim_{J \rightarrow \infty} \gamma^* = 1 - \frac{\alpha \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right) \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}{\alpha^2 \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right)^2 \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right)} = 1 - \frac{\left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right)}{\alpha \left(\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2} \right) \left(\sigma_x^2 + \sigma_y^2 + \frac{\sigma_0^2}{J} \right)} = 1 - \frac{1}{\alpha \frac{\frac{\sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}}{\frac{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}}} = 1 - \frac{\beta}{\alpha}$.

Next, we can express $\hat{\theta}^*$ for large crowds. To begin with, observe that $\lim_{J \rightarrow \infty} \bar{f}^1 = \lim_{J \rightarrow \infty} w f^a + (1 - w) \bar{f}^0 + \frac{1}{J} \sum_{j=1}^J \varepsilon_j^1 = w f^a + (1 - w) \bar{f}^0$. Next:

$$\begin{aligned} \lim_{J \rightarrow \infty} \hat{\theta}^* &= \lim_{J \rightarrow \infty} \bar{f}^1 + \gamma^* (\bar{f}^0 - \bar{f}^1) = \lim_{J \rightarrow \infty} (1 - \gamma^*) \bar{f}^1 + \gamma^* \bar{f}^0 \\ &= (1 - \tilde{\gamma})(w f^a + (1 - w) \bar{f}^0) + \tilde{\gamma} \bar{f}^0 \quad (\text{since } \lim_{J \rightarrow \infty} \gamma^* = \tilde{\gamma} \text{ and } \lim_{J \rightarrow \infty} \bar{f}^1 = w f^a + (1 - w) \bar{f}^0) \\ &= \frac{\beta}{\alpha} (w f^a + (1 - w) \bar{f}^0) + \left(1 - \frac{\beta}{\alpha} \right) \bar{f}^0 \quad (\text{since } \tilde{\gamma} = 1 - \frac{\beta}{\alpha}) \\ &= \frac{\beta}{\alpha} (\alpha w^* f^a + (1 - \alpha w^*) \bar{f}^0) + \left(1 - \frac{\beta}{\alpha} \right) \bar{f}^0 \quad (\text{since } w = \alpha w^*) \\ &= \beta w^* f^a + \frac{\beta}{\alpha} \bar{f}^0 - \beta w^* \bar{f}^0 + \bar{f}^0 - \frac{\beta}{\alpha} \bar{f}^0 = \beta w^* f^a + (1 - \beta w^*) \bar{f}^0 = \theta^*, \end{aligned}$$

i.e., $\hat{\theta}^*$ is a consistent estimator of θ^* , which completes the proof. $\square$

EC.2.6 Proof of Proposition 6

First, we calculate the expected squared forecast error of the optimally pivoted forecast:

$$\begin{aligned} \mathcal{E}(\hat{\theta}^*) &= \mathcal{E}(\bar{f}^1 + \gamma^* (\bar{f}^0 - \bar{f}^1)) = \mathcal{E}((1 - \gamma^*) \bar{f}^1 + \gamma^* \bar{f}^0) = \mathbb{E} \left[((1 - \gamma^*) \bar{f}^1 + \gamma^* \bar{f}^0 - \theta)^2 \middle| \theta \right] \\ &= \mathbb{E} \left[((1 - \gamma^*) (\bar{f}^1 - \theta))^2 \middle| \theta \right] + \mathbb{E} \left[(\gamma^* (\bar{f}^0 - \theta))^2 \middle| \theta \right] + \mathbb{E} \left[2(1 - \gamma^*) \gamma^* (\bar{f}^1 - \theta) (\bar{f}^0 - \theta) \middle| \theta \right] \\ &= (1 - \gamma^*)^2 \mathbb{E} \left[\left(w(\theta + s + x) + (1 - w) \left(\theta + s + y + \frac{\sum_{j=1}^J \varepsilon_j^0}{J} \right) + \frac{\sum_{j=1}^J \varepsilon_j^1}{J} - \theta \right)^2 \middle| \theta \right] \\ &\quad + \gamma^{*2} \mathbb{E} \left[\left(\theta + s + y + \frac{\sum_{j=1}^J \varepsilon_j^0}{J} - \theta \right)^2 \middle| \theta \right] \\ &\quad + 2\gamma^*(1 - \gamma^*) \cdot \mathbb{E} \left[\left(w(\theta + s + x) + (1 - w) \left(\theta + s + y + \frac{\sum_{j=1}^J \varepsilon_j^0}{J} \right) + \frac{\sum_{j=1}^J \varepsilon_j^1}{J} - \theta \right) \left(\theta + s + y + \frac{\sum_{j=1}^J \varepsilon_j^0}{J} - \theta \right) \middle| \theta \right] \\ &= (1 - \gamma^*)^2 \mathbb{E} \left[\left(s + wx + (1 - w) \left(y + \frac{\sum_{j=1}^J \varepsilon_j^0}{J} \right) + \frac{\sum_{j=1}^J \varepsilon_j^1}{J} \right)^2 \middle| \theta \right] + \gamma^{*2} \mathbb{E} \left[\left(s + y + \frac{\sum_{j=1}^J \varepsilon_j^0}{J} \right)^2 \middle| \theta \right] \end{aligned}$$

$$\begin{aligned}
& + 2\gamma^*(1-\gamma^*)\mathbb{E} \left[\left(s + wx + (1-w) \left(y + \frac{\sum_{j=1}^J \varepsilon_j^0}{J} \right) + \frac{\sum_{j=1}^J \varepsilon_j^1}{J} \right) \left(s + y + \frac{\sum_{j=1}^J \varepsilon_j^0}{J} \right) \middle| \theta \right] \\
& = ((1-\gamma^*)^2 + \gamma^{*2} + 2\gamma^*(1-\gamma^*)) \cdot \sigma_s^2 \\
& + (w(1-\gamma^*))^2 \cdot \sigma_x^2 + ((1-\gamma^*)^2(1-w)^2 + \gamma^{*2} + 2\gamma^*(1-\gamma^*)(1-w)) \cdot \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) + (1-\gamma^*)^2 \cdot \frac{\sigma_1^2}{J} \\
& = \sigma_s^2 + (w(1-\gamma^*))^2 \cdot \sigma_x^2 + (1-w(1-\gamma^*))^2 \cdot \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) + (1-\gamma^*)^2 \cdot \frac{\sigma_1^2}{J}.
\end{aligned}$$

Using the value of $\mathcal{E}(\bar{f}^1)$ from the proof of Proposition 3, we can show:

$$\begin{aligned}
\mathcal{E}(\bar{f}^1) - \mathcal{E}(\hat{\theta}^*) & = \sigma_s^2 + w^2\sigma_x^2 + (1-w)^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) + \frac{\sigma_1^2}{J} \\
& \quad - \sigma_s^2 - (w(1-\gamma^*))^2 \sigma_x^2 - (1-w(1-\gamma^*))^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) - (1-\gamma^*)^2 \frac{\sigma_1^2}{J} \\
& = w^2\sigma_x^2 + (1-w)^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) + \frac{\sigma_1^2}{J} \\
& \quad - (w(1-\gamma^*))^2 \sigma_x^2 - (1-w(1-\gamma^*))^2 \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) - (1-\gamma^*)^2 \frac{\sigma_1^2}{J} \\
& = \left(w^2 - (w(1-\gamma^*))^2 \right) \sigma_x^2 + ((1-w)^2 - (1-w(1-\gamma^*))^2) \left(\sigma_y^2 + \frac{\sigma_0^2}{J} \right) + (1 - (1-\gamma^*)^2) \frac{\sigma_1^2}{J} \\
\Rightarrow \lim_{J \rightarrow \infty} \mathcal{E}(\bar{f}^1) - \mathcal{E}(\hat{\theta}^*) & = \left(w^2 - (w(1-\tilde{\gamma}))^2 \right) \sigma_x^2 + ((1-w)^2 - (1-w(1-\tilde{\gamma}))^2) \sigma_y^2 \\
& = \left(w^2 - \left(\frac{\beta w}{\alpha} \right)^2 \right) \sigma_x^2 + \left((1-w)^2 - \left(1 - \frac{\beta w}{\alpha} \right)^2 \right) \sigma_y^2 \quad (\text{since } \tilde{\gamma} = 1 - \frac{\beta}{\alpha}) \\
& = ((\alpha w^*)^2 - (\beta w^*)^2) \sigma_x^2 + ((1-\alpha w^*)^2 - (1-\beta w^*)^2) \sigma_y^2 \quad (\text{since } w = \alpha w^*) \\
& = (\alpha + \beta)(\alpha - \beta)w^{*2}\sigma_x^2 + (2 - (\alpha + \beta)w^*)(\beta - \alpha)w^*\sigma_y^2 \\
& = (\alpha - \beta)w^*(\sigma_x^2 + \sigma_y^2) \left((\alpha + \beta)w^* \frac{\sigma_x^2}{\sigma_x^2 + \sigma_y^2} - (2 - (\alpha + \beta)w^*) \frac{\sigma_y^2}{\sigma_x^2 + \sigma_y^2} \right) \\
& = (\alpha - \beta)w^*(\sigma_x^2 + \sigma_y^2) ((\alpha + \beta)w^*(1 - \beta w^*) - (2 - (\alpha + \beta)w^*)\beta w^*) \\
& \quad \left(\text{since } \lim_{J \rightarrow \infty} \beta = \frac{\frac{\sigma_y^2}{\sigma_y^2 + \sigma_0^2}}{\frac{\sigma_x^2 + \sigma_y^2}{\sigma_x^2 + \sigma_y^2 + \sigma_0^2}} = \frac{1}{w^*} \frac{\sigma_y^2}{\sigma_x^2 + \sigma_y^2}, \text{ which yields } \frac{\sigma_y^2}{\sigma_x^2 + \sigma_y^2} = \beta w^* \right) \\
& = (\alpha - \beta)w^*(\sigma_x^2 + \sigma_y^2) ((\alpha + \beta)w^* - (\alpha + \beta)w^*\beta w^* - 2\beta w^* + (\alpha + \beta)w^*\beta w^*) \\
& = (\alpha - \beta)w^*(\sigma_x^2 + \sigma_y^2) ((\alpha + \beta)w^* - 2\beta w^*) = (\alpha - \beta)^2 w^{*2} (\sigma_x^2 + \sigma_y^2),
\end{aligned}$$

and that completes the proof. $\square$

EC.2.7 Proof of Proposition 7

Given historical data $\{(X_i, Y_i)\}_{i=1}^N$, we define the empirical in-sample loss function as $\hat{L}(\gamma) \stackrel{\text{def}}{=} \sum_{i=1}^N (Y_i - \gamma X_i)^2$. $\hat{L}(\gamma)$ is a quadratic function of γ , and since the leading coefficient $\sum_{i=1}^N X_i^2 > 0$, we can see that $\hat{L}(\gamma)$ is convex in γ . Expanding the function, we get $\hat{L}(\gamma) = \sum_{i=1}^N Y_i^2 - 2\gamma \sum_{i=1}^N X_i Y_i +$

$\gamma^2 \sum_{i=1}^N X_i^2$ which is continuous and differentiable in γ . Differentiating with respect to γ yields $\frac{\partial \hat{L}(\gamma)}{\partial \gamma} = -2 \sum_{i=1}^N X_i Y_i + 2\gamma \sum_{i=1}^N X_i^2$. Setting the derivative equal to zero gives the following unique unconstrained minimizer of (14): $\hat{\gamma}_0 = \sum_{i=1}^N X_i Y_i / \sum_{i=1}^N X_i^2$. Since $\hat{L}(\gamma)$ is convex, the constrained minimizer coincides with the unconstrained minimizer whenever $\hat{\gamma}_0 \leq 1$, and otherwise equals the boundary point $\gamma = 1$. Therefore, the solution to the constrained minimization problem is given by $\hat{\gamma} = \min \left\{ \sum_{i=1}^N X_i Y_i / \sum_{i=1}^N X_i^2, 1 \right\}$, which completes the proof. $\square$

EC.2.8 Proof of Proposition 8

We can write the objective function in (17) as the empirical risk with penalization from using a pivoting factor of γ : $\hat{L}_\lambda(\gamma) \stackrel{\text{def}}{=} \sum_{i=1}^N (Y_i - \gamma X_i)^2 + \lambda|\gamma| = \sum_{i=1}^N Y_i^2 - 2\gamma \sum_{i=1}^N X_i Y_i + \gamma^2 \sum_{i=1}^N X_i^2 = Q\gamma^2 - 2S\gamma + \lambda|\gamma| + \text{const}$. Observe that $Q\gamma^2 - 2S\gamma$ is convex in γ for $Q > 0$, and $\lambda|\gamma|$ is also convex in γ ; therefore $\hat{L}_\lambda(\gamma)$ is convex in γ . As a result, $\hat{\gamma}_{\lambda_0}$ minimizes $\hat{L}_\lambda$ over $\mathbb{R}$ if and only if $0 \in \partial \hat{L}_\lambda(\hat{\gamma}_{\lambda_0})$, where ∂ denotes the subdifferential (Boyd and Vandenberghe 2004). For $\gamma \neq 0$, we have $\partial \hat{L}_\lambda(\gamma) = \{2Q\gamma - 2S + \lambda \text{sgn}(\gamma)\}$. If $\hat{\gamma}_{\lambda_0} > 0$, then $\text{sgn}(\hat{\gamma}_{\lambda_0}) = 1$, and the first-order condition becomes $2Q\hat{\gamma}_{\lambda_0} - 2S + \lambda = 0$, giving us $\hat{\gamma}_{\lambda_0} = (S - \lambda/2)/Q$. This is feasible only if $S > \lambda/2$. On the other hand if $\hat{\gamma}_{\lambda_0} < 0$, then $\text{sgn}(\hat{\gamma}_{\lambda_0}) = -1$, and the first-order condition becomes $2Q\hat{\gamma}_{\lambda_0} - 2S - \lambda = 0$, giving us now $\hat{\gamma}_{\lambda_0} = (S + \lambda/2)/Q$. This is feasible only if $S < -\lambda/2$. If $|S| \leq \lambda/2$, then $0 \in [-2S - \lambda, -2S + \lambda] = \partial F(0)$, and so $\hat{\gamma}_{\lambda_0} = 0$ is optimal. Combining these three cases, we get

$$\begin{aligned} \hat{\gamma}_{\lambda_0} &= \begin{cases} (S - \lambda/2)/Q, & \text{if } S > \lambda/2; \\ (S + \lambda/2)/Q, & \text{if } S < -\lambda/2; \\ 0, & \text{otherwise.} \end{cases} \\ &= \text{soft}(S, \lambda/2)/Q. \end{aligned}$$

Finally, since $\hat{L}_\lambda$ is convex in γ , if the global minima $\hat{\gamma}_{\lambda_0}$ is greater than 1, the constrained regularized minimization problem is solved at $\hat{\gamma}_\lambda = \min \{\hat{\gamma}_{\lambda_0}, 1\}$. That completes the proof. $\square$

EC.3 Statistical Guarantees of Data-Driven Pivoting

Proposition EC.1 provides a finite-sample learning guarantee on $\hat{\gamma}$, demonstrating how quickly $\hat{\gamma}$ concentrates around γ^* as the number of historical tasks increases.

PROPOSITION EC.1. *Assume that the N historical tasks are i.i.d. draws from the same environment and that $|X_i| \leq B$ and $|Y_i| \leq B$ almost surely for some $B > 0$, for all $i \in \{1, \dots, N\}$. Assume further the non-degeneracy condition $\mathbb{E}[X^2] \geq c > 0$. For any $\delta \in (0, 1)$, if $c \geq 2B^2 \sqrt{\frac{\log(4/\delta)}{2N}} > 0$, then with probability at least $1 - \delta$,*

$$|\hat{\gamma} - \gamma^*| \leq \left(\frac{2}{c} + \frac{2B^2}{c^2} \right) B^2 \sqrt{\frac{\log(4/\delta)}{2N}} = \mathcal{O}(N^{-1/2}). \quad (\text{EC.18})$$

In Proposition EC.1, the assumption of boundedness of X and Y is mild because forecasts and outcomes are typically recorded on finite scales (or can be clipped without materially affecting performance). Here, the non-degeneracy condition $\mathbb{E}[X^2] \geq c > 0$ at a population level rules out the uninformative case in which the crowd rarely revises after seeing AI advice, and the data contain too little variation to identify how much pivoting is beneficial, resulting in any estimator of γ^* being unstable. It delivers a simple and actionable message: learning how much to pivot is feasible precisely when the historical environment contains enough variation in how the crowd responds to AI. When adjustments are frequent and sizable, $\hat{\gamma}$ concentrates around γ^* at the familiar $\mathcal{O}(N^{-1/2})$ rate as the number of historical problems grows. When revisions are typically small, $\hat{\gamma}$ can be unstable in finite samples, making aggressive pivoting difficult to justify from limited data.

Next, we evaluate $\hat{\gamma}$ through the lens of population risk. Let (X, Y) denote a generic draw from the (stable) forecasting environment and define the expected squared forecast error as a function of the pivoting factor, $L(\gamma) \stackrel{\text{def}}{=} \mathbb{E}[(Y - \gamma X)^2]$. Because the DM ultimately cares about forecast accuracy rather than the pivoting factor itself, the next proposition translates this concentration result into an explicit out-of-sample performance guarantee by bounding the regret that arises from using the empirically learned pivoting factor vis-à-vis knowing γ^* , i.e., $L(\hat{\gamma}) - L(\gamma^*)$.

PROPOSITION EC.2. *Under the assumptions of Proposition EC.1, with probability at least $1 - \delta$,*

$$L(\hat{\gamma}) - L(\gamma^*) \leq \mathbb{E}[X^2] \left(\frac{2}{c} + \frac{2B^2}{c^2} \right)^2 \frac{B^4 \log(4/\delta)}{2N} = \mathcal{O}(N^{-1}). \quad (\text{EC.19})$$

Whereas Proposition EC.1 shows that the estimation error of our empirical pivoting factor shrinks at the usual parametric rate $\mathcal{O}(N^{-1/2})$, Proposition EC.2 shows that its out-of-sample regret decays even faster at $\mathcal{O}(N^{-1})$. For the DM, this is reassuring: even if $\hat{\gamma}$ is learned from a finite historical sample, its performance approaches that of γ^* as the amount of relevant historical data grows. Clearly, having a large number of historical forecasts is helpful for the DM, but if this is not available, the DM needs to take into consideration the instabilities arising from small samples when estimating the pivoting factor using regularization, as described in §6.1.

We extend the asymptotic analysis of the empirical pivoting process by taking into account the regularization process. Proposition EC.3 quantifies this tradeoff by bounding the amount of shrinkage induced by λ and relating the resulting error of $\hat{\gamma}_\lambda$ to that of the unregularized $\hat{\gamma}$.

PROPOSITION EC.3. *Assuming $Q > 0$, the absolute difference between the estimators $\hat{\gamma}$ and $\hat{\gamma}_\lambda$ can be bounded as*

$$|\hat{\gamma}_\lambda - \gamma^*| \leq |\hat{\gamma} - \gamma^*| + \frac{\lambda}{2Q}. \quad (\text{EC.20})$$

Proposition EC.3 explicitly characterizes the bias-variance tradeoff in the regularization process. The ℓ_1 -penalty shrinks the unregularized slope toward zero, and the resulting shrinkage bias is tightly controlled (at most $\lambda/(2Q)$). In exchange, regularization improves finite-sample stability, especially when Q is small—precisely the regime in which unregularized estimation is ill-conditioned because the crowd’s revisions after seeing AI advice are limited. Similarly, larger values of λ safeguard the out-of-sample forecasting accuracy of $\hat{\theta}_\lambda$ against spurious correlations in the historical forecasts, but also concede a higher possibility of shrinkage-biased estimates of γ^* .

We conclude the discussion on the bias-variance tradeoff by translating the shrinkage bias in the regularized pivoting factor $\hat{\gamma}_\lambda$ into an explicit out-of-sample performance guarantee by bounding the regret that arises exclusively from regularization, i.e., $L(\hat{\gamma}_\lambda) - L(\hat{\gamma})$, in the following proposition.

PROPOSITION EC.4. *Under the assumptions of Proposition EC.1, if $Q > 0$, then with probability at least $1 - \delta$,*

$$L(\hat{\gamma}_\lambda) - L(\hat{\gamma}) \leq \mathbb{E}[X^2] \left(\frac{\lambda}{Q} \left(\frac{2}{c} + \frac{2B^2}{c^2} \right) B^2 \sqrt{\frac{\log(4/\delta)}{2N}} + \frac{\lambda^2}{4Q^2} \right). \quad (\text{EC.21})$$

Proposition EC.4 demonstrates that upon maintaining a fixed N and Q for a given forecasting context, the excess risk that can be attributed specifically to shrinkage bias caused by regularization scales quadratically in λ . This should serve as a precaution against using very high values of the λ ; even though the absolute difference $|\hat{\gamma}_\lambda - \gamma^*|$ grows linearly at a rate of $\mathcal{O}(\lambda)$, Proposition EC.4 reveals that the excess risk taken on by the DM in choosing to stabilize the estimated pivoting factor grows at a faster rate of $\mathcal{O}(\lambda^2)$. Proofs of Propositions EC.1–EC.4 are provided in Appendix A.

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A Appendix

A.1 Proof of Proposition EC.1

Define the finite-sample means of XY and X^2 as $\hat{\mu}_1 \stackrel{\text{def}}{=} \frac{1}{N} \sum_{i=1}^N X_i Y_i$ and $\hat{\mu}_2 \stackrel{\text{def}}{=} \frac{1}{N} \sum_{i=1}^N X_i^2$, respectively, and similarly define their corresponding population means as $\mu_1 \stackrel{\text{def}}{=} \mathbb{E}[XY]$ and $\mu_2 \stackrel{\text{def}}{=} \mathbb{E}[X^2]$, respectively. Note from Proposition 7 that the unconstrained empirical minimizer of (14) is $\hat{\gamma}_0 = \hat{\mu}_1 / \hat{\mu}_2$ given $\hat{\mu}_2 > 0$, and the constrained empirical minimizer is the projection $\hat{\gamma} = \hat{\gamma}_0 \wedge 1$. Likewise, the unconstrained population minimizer of $L(\gamma)$ is $\gamma_0^* = \mu_1 / \mu_2$ whenever $\mu_2 > 0$, and the constrained population minimizer is $\gamma^* = \gamma_0^* \wedge 1$.

Given that $|X_i| \leq B$ and $|Y_i| \leq B$, we have $|X_i Y_i| \leq B^2$, i.e. $-B^2 \leq X_i Y_i \leq B^2$, and $0 \leq X_i^2 \leq B^2$, almost surely for all $i \in \{1, \dots, N\}$. Next, define $t \stackrel{\text{def}}{=} B^2 \sqrt{\frac{\log(4/\delta)}{2N}} > 0$ for any $\delta \in (0, 1)$. Note that using the condition $c \geq 2B^2 \sqrt{\frac{\log(4/\delta)}{2N}} > 0$, we have $t \leq c/2$. Now using Hoeffding's inequality (Hoeffding 1963), we have for all $t > 0$, $\Pr(|\hat{\mu}_1 - \mu_1| \geq t) = \Pr\left(\left|\frac{1}{N} \sum_{i=1}^N X_i Y_i - \mathbb{E}[XY]\right| \geq t\right) \leq 2 \exp\left(-2N^2 t^2 / \sum_{i=1}^N (B^2 - (-B^2))^2\right) = 2 \exp\left(-\frac{2N^2 t^2}{4NB^4}\right) = 2 \exp\left(-\frac{Nt^2}{2B^4}\right)$ and similarly, $\Pr(|\hat{\mu}_2 - \mu_2| \geq t) = \Pr\left(\left|\frac{1}{N} \sum_{i=1}^N X_i^2 - \mathbb{E}[X^2]\right| \geq t\right) \leq 2 \exp\left(-2N^2 t^2 / \sum_{i=1}^N (B^2 - 0)^2\right) = 2 \exp\left(-\frac{2N^2 t^2}{NB^4}\right) = 2 \exp\left(-\frac{2Nt^2}{B^4}\right)$. Plugging in $t = B^2 \sqrt{\frac{\log(4/\delta)}{2N}}$ for any $\delta \in (0, 1)$, we get

$$\begin{aligned} \Pr(|\hat{\mu}_1 - \mu_1| \geq t) &\leq 2 \exp\left(-\frac{Nt^2}{2B^4}\right) = 2 \exp\left(-\frac{NB^4 \log(4/\delta)/(2N)}{2B^4}\right) = 2 \exp\left(-\frac{\log(4/\delta)}{4}\right) \\ &= 2 \exp\left(\frac{1}{4} \log(\delta/4)\right) \leq 2 \exp(\log(\delta/4)) = \delta/2, \end{aligned}$$

and also

$$\Pr(|\hat{\mu}_2 - \mu_2| \geq t) \leq 2 \exp\left(-\frac{2Nt^2}{B^4}\right) = 2 \exp\left(-\frac{2NB^4 \log(4/\delta)/(2N)}{B^4}\right) = 2 \exp(\log(\delta/4)) = \delta/2.$$

This shows us that both probabilities are at most $\delta/2$. Then, by a union bound, we define the event $\mathcal{U} \stackrel{\text{def}}{=} \{|\hat{\mu}_1 - \mu_1| \leq t\} \cap \{|\hat{\mu}_2 - \mu_2| \leq t\}$ which satisfies $\Pr(\mathcal{U}) \geq 1 - \delta$. Upon the event $\mathcal{U}$, we have that $\hat{\mu}_2 \geq \mu_2 - t \geq c - t \geq c/2 > 0$, since $t \leq c/2$ and $c > 0$. Consequently, $\hat{\gamma}_0$ is well-defined and $1/\hat{\mu}_2 \leq 2/c$ on the event $\mathcal{U}$. Furthermore, on the event $\mathcal{U}$,

$$\begin{aligned} |\hat{\gamma}_0 - \gamma_0^*| &= \left| \frac{\hat{\mu}_1}{\hat{\mu}_2} - \frac{\mu_1}{\mu_2} \right| = \left| \frac{\hat{\mu}_1 \mu_2 - \mu_1 \hat{\mu}_2}{\hat{\mu}_2 \mu_2} \right| = \left| \frac{\hat{\mu}_1 \mu_2 - \mu_1 \mu_2 + \mu_1 \mu_2 - \mu_1 \hat{\mu}_2}{\hat{\mu}_2 \mu_2} \right| = \left| \frac{\hat{\mu}_1 - \mu_1}{\hat{\mu}_2} + \frac{\mu_1(\mu_2 - \hat{\mu}_2)}{\hat{\mu}_2 \mu_2} \right| \\ &\leq \left| \frac{\hat{\mu}_1 - \mu_1}{\hat{\mu}_2} \right| + \left| \frac{\mu_1(\mu_2 - \hat{\mu}_2)}{\hat{\mu}_2 \mu_2} \right| \leq \frac{|\hat{\mu}_1 - \mu_1|}{\hat{\mu}_2} + \frac{|\mu_1|}{\mu_2} \cdot \frac{|\mu_2 - \hat{\mu}_2|}{\hat{\mu}_2} \quad (\text{since } \hat{\mu}_2 \geq c/2 > 0 \text{ and } \mu_2 > c > 0). \end{aligned}$$

Using $|\hat{\mu}_1 - \mu_1| \leq t$, $|\hat{\mu}_2 - \mu_2| \leq t$, $\hat{\mu}_2 \geq c/2$, $\mu_2 \geq c$, and $|\mu_1| = |\mathbb{E}[XY]| \leq \mathbb{E}[|X||Y|] \leq B^2$ upon the event $\mathcal{U}$, we obtain $|\hat{\gamma}_0 - \gamma_0^*| \leq \frac{t}{c/2} + \frac{B^2}{c} \cdot \frac{t}{c/2} = \left(\frac{2}{c} + \frac{2B^2}{c^2}\right) t$. Also, given that the projection $\mathbb{R} \rightarrow (-\infty, 1]$ using the function $\min\{z, 1\}$ for all $z \in \mathbb{R}$ is a non-expansive mapping (i.e., it cannot make values further apart) (Boyd and Vandenberghe 2004), we have that $|\hat{\gamma} - \gamma^*| \leq |\hat{\gamma}_0 - \gamma_0^*| \leq \left(\frac{2}{c} + \frac{2B^2}{c^2}\right) B^2 \sqrt{\frac{\log(4/\delta)}{2N}}$. Finally, since the event $\mathcal{U}$ occurs with a probability at least $1 - \delta$, and since the event $\mathcal{U}$ is a sufficient condition for the above inequality to hold, we can claim that the inequality $|\hat{\gamma} - \gamma^*| \leq \left(\frac{2}{c} + \frac{2B^2}{c^2}\right) B^2 \sqrt{\frac{\log(4/\delta)}{2N}}$ also holds with a probability at least $1 - \delta$, which completes the proof. $\square$

A.2 Proof of Proposition EC.2

First, recall that $L(\gamma) = \mathbb{E}[(Y - \gamma X)^2]$, $\mu_1 = \mathbb{E}[XY]$, and $\mu_2 = \mathbb{E}[X^2]$. Completing the square,

$$\begin{aligned} L(\gamma) &= \mathbb{E}[Y^2] - 2\gamma\mathbb{E}[XY] + \gamma^2\mathbb{E}[X^2] = \mathbb{E}[Y^2] - 2\gamma\mu_1 + \gamma^2\mu_2 = \mathbb{E}[Y^2] - \frac{\mu_1^2}{\mu_2} + \left(\mu_2\frac{\mu_1^2}{\mu_2^2} - 2\mu_2\gamma\frac{\mu_1}{\mu_2} + \gamma^2\mu_2\right) \\ &= \mathbb{E}[Y^2] - \frac{\mu_1^2}{\mu_2} + \mu_2\left(\gamma - \frac{\mu_1}{\mu_2}\right)^2 = \text{const} + \mu_2(\gamma - \gamma_0^*)^2, \end{aligned}$$

where $\gamma_0^* = \mu_1/\mu_2$ is the unconstrained minimizer of population risk. Next, note that under a Bayesian model of the forecasting environment described in §3–5, $\gamma^* < 1$ holds. Since this also implies that $\gamma_0^* < 1$, the constrained minimizer is $\gamma^* = \gamma_0^*$, and therefore $L(\hat{\gamma}) - L(\gamma^*) = L(\hat{\gamma}) - L(\gamma_0^*) = \mu_2(\hat{\gamma} - \gamma_0^*)^2 = \mathbb{E}[X^2] \cdot (\hat{\gamma} - \gamma^*)^2$. Combining this result with equation (EC.18) in Proposition EC.1 which establishes a bound on the absolute difference between the constrained empirical risk minimizer $\hat{\gamma}$ and the constrained population risk minimizer γ^* , we can show that with probability at least $1 - \delta$, the regret from using the empirically estimated pivoting factor will be less than $\mathbb{E}[X^2] \left(\frac{2}{c} + \frac{2B^2}{c^2}\right)^2 \frac{B^4 \log(4/\delta)}{2N}$. That completes the proof. $\square$

A.3 Proof of Proposition EC.3

For any $s \in \mathbb{R}$ and $a \geq 0$, $|\text{soft}(s, a) - s| \leq a$, i.e., soft-thresholding shrinks by at most a , and since $Q > 0$, we can establish bounds upon the absolute difference between the unconstrained estimators: $|\hat{\gamma}_{\lambda_0} - \hat{\gamma}_0| = |\text{soft}(S, \lambda/2)/Q - S/Q| = |\text{soft}(S, \lambda/2) - S|/Q \leq \lambda/(2Q)$. Next, given that the projection $\mathbb{R} \rightarrow (-\infty, 1]$ using the function $\min\{z, 1\}$ for all $z \in \mathbb{R}$ is a non-expansive mapping (i.e., it cannot make values further apart) (Boyd and Vandenberghe 2004), we can extend the results to the solutions of the constrained empirical risk minimization problems. Therefore, $|\hat{\gamma}_\lambda - \hat{\gamma}| \leq |\hat{\gamma}_{\lambda_0} - \hat{\gamma}_0| \leq \lambda/(2Q)$. Finally, we introduce the optimal pivoting factor γ^* , and using the triangle inequality, $|\hat{\gamma}_\lambda - \gamma^*| \leq |\hat{\gamma} - \gamma^*| + |\hat{\gamma}_\lambda - \hat{\gamma}| \leq |\hat{\gamma} - \gamma^*| + \lambda/(2Q)$, which completes the proof. $\square$

A.4 Proof of Proposition EC.4

From the proof of Proposition EC.2, we have $L(\gamma) = \text{const} + \mu_2(\gamma - \gamma_0^*)^2 = \text{const} + \mu_2(\gamma - \gamma^*)^2$, since $\gamma^* = \gamma_0^* < 1$ under our model conditions. To compare the excess risk due to regularization, we can calculate

$$\begin{aligned} L(\hat{\gamma}_\lambda) - L(\gamma^*) &= \mu_2(\hat{\gamma}_\lambda - \gamma^*)^2 = \mu_2((\hat{\gamma}_\lambda - \hat{\gamma}) + (\hat{\gamma} - \gamma^*))^2 = \mu_2(\hat{\gamma} - \gamma^*)^2 + \mu_2(2(\hat{\gamma}_\lambda - \hat{\gamma})(\hat{\gamma} - \gamma^*) + (\hat{\gamma}_\lambda - \hat{\gamma})^2) \\ &= (L(\hat{\gamma}) - L(\gamma^*)) + \mu_2(2(\hat{\gamma}_\lambda - \hat{\gamma})(\hat{\gamma} - \gamma^*) + (\hat{\gamma}_\lambda - \hat{\gamma})^2) \\ \Rightarrow L(\hat{\gamma}_\lambda) - L(\hat{\gamma}) &= \mu_2(2(\hat{\gamma}_\lambda - \hat{\gamma})(\hat{\gamma} - \gamma^*) + (\hat{\gamma}_\lambda - \hat{\gamma})^2) \leq 2\mu_2|\hat{\gamma}_\lambda - \hat{\gamma}||\hat{\gamma} - \gamma^*| + \mu_2(\hat{\gamma}_\lambda - \hat{\gamma})^2 \\ &\leq 2\mu_2\frac{\lambda}{2Q}|\hat{\gamma} - \gamma^*| + \mu_2\frac{\lambda^2}{4Q^2}, \end{aligned}$$

using the intermediate result from the proof of Proposition EC.3. Finally, using the bound on $|\hat{\gamma} - \gamma^*|$ described in Proposition EC.1, we can claim that with probability at least $1 - \delta$, $L(\hat{\gamma}_\lambda) - L(\hat{\gamma}) \leq \mathbb{E}[X^2] \left(\frac{\lambda}{Q} \left(\frac{2}{c} + \frac{2B^2}{c^2}\right) B^2 \sqrt{\frac{\log(4/\delta)}{2N}} + \frac{\lambda^2}{4Q^2}\right)$, which completes the proof. $\square$